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Robust treatment assignment with uncertainty-aware causal forests: Joint optimization of accuracy and estimation uncertainty

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ABSTRACT

Causal treatment decision-making methods commonly aim at point-estimate accuracy of causal effect estimation and overlook the estimation uncertainty, leading to overconfident policies that may fail to generalize reliably in real-world settings. This study introduces an *Uncertainty-Aware Causal Forest* framework that explicitly trains causal forests to minimize finite-sample estimation uncertainty. Our approach achieves this goal by embedding infinitesimal-jackknife-based confidence interval (CI) width directly into a joint uncertainty–accuracy learning formulation. Specifically, the framework couples Gaussian-process Bayesian optimization-for inferring a joint probabilistic model of predictive error and CI width and for adaptively exploring the hyperparameter space-with a fully nested, honest cross-validation architecture. This integration ensures hyperparameter selection is guided by out-of-sample uncertainty behavior rather than idiosyncrasies of the training data, yielding well-calibrated and robust conditional average treatment effect (CATE) estimates. Across one synthetic and two real-world benchmark datasets, the proposed framework reduces test CI width by more than an order of magnitude compared with Generalized Random Forests and Double Machine Learning, while maintaining or improving predictive accuracy, achieving near-nominal uncertainty coverage, and substantially shrinking train-test discrepancies. A dual-threshold targeting rule, supplemented by a lower-confidence-bound ranking strategy, translates these gains into actionable treatment decisions by prioritizing individuals with high estimated effects and low uncertainty. By coupling uncertainty minimization with practical decision rules, the proposed framework offers a transparent, statistically robust, and uncertainty-aware foundation for expert systems in healthcare, marketing, and policy applications.

1. Introduction

In recent years, there has been a growing focus on understanding causal relationships across diverse fields, driven by advances in machine learning (ML) (Chen et al., 2022; Jordan & Mitchell, 2015; Ma & Sun, 2020; Sanchez et al., 2022). Among these ML methods, causal forests have emerged as a particularly powerful tool for improving decision-making by offering personalized treatment effects that inform interventions across settings, such as healthcare and education (Hattab et al., 2024; Venkatasubramaniam et al., 2023; Webbink, 2005). By estimating the Conditional Average Treatment Effect (CATE), causal forests enable assessment of treatment impacts at individual or subgroup levels, thereby tailoring interventions based on observed characteristics (Kaddour et al., 2022).

Among these methods, causal forests have emerged as a particularly powerful tool for estimating heterogeneous treatment effects and guiding individualized decisions (Athey & Wager, 2019; Wager & Athey, 2018). By estimating the CATE, causal forests enable assessment of treatment impacts at individual or subgroup levels, thereby tailoring interventions based on observed characteristics (Kaddour et al., 2022). Pioneering contributions by Wager and Athey (Wager & Athey, 2018) adapted random forests to causal inference, providing consistent non-parametric estimation of heterogeneous effects. Athey and Wager (Athey & Wager, 2019) subsequently demonstrated their practical utility in education through the National Study of Learning Mindsets (ATHEY et al., 2019). Subsequent applications extended into domains such as marketing (Herhausen et al., 2024; Langen & Huber, 2023), energy (Li et al., 2025; Mao et al., 2025), and public policy.

Parallel to these developments, recent research has advanced causal inference toward richer data environments and adaptive decision contexts. Examples include causal discovery and reasoning with multimodal data for root-cause analysis in complex systems (Chen et al., 2025), causal partitioning and structure learning through graph-based edge cutting (Zhang et al., 2024), and causal-intervention-based visual

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navigation and feature selection for explainable artificial intelligence (Tan, 2025; Zhao et al., 2023). These studies exemplify a broader shift toward causally-informed representation learning, where graph-based or intervention-aware models enhance interpretability and transferability.

Simultaneously, the field has witnessed rapid progress in deep causal estimators and causal meta-learners, such as DragonNet, TARNet, and CEVAE, which leverage neural architectures for end-to-end treatment effect estimation (Louizos et al., 2017; Shalit et al., 2017; Shi et al., 2019). Reinforcement-learning-based causal policy optimization frameworks have also been developed to adaptively learn optimal treatment rules under uncertainty (Deng et al., 2023). Collectively, these contributions underscore an ongoing convergence of causal inference, deep learning, and decision theory toward unified, uncertainty-aware learning paradigms.

Beyond causal inference, the broader intelligent-systems literature has increasingly emphasized the role of uncertainty-aware learning and Bayesian regularization in enhancing model robustness and interpretability. For example, recent studies have introduced robust model predictive control for fractional-order systems with uncertainty (Arbi, 2024), Bayesian regularization methods for nonlinear dynamic systems (Sabir et al., 2023, 2025), and hybrid neural architectures combining radial basis and sigmoid functions for biomedical prediction tasks (Sabir et al., 2024). Similarly, machine and deep learning models have been successfully applied in financial prediction tasks under uncertainty, such as bankruptcy forecasting (Hamdi et al., 2024). Although these approaches differ in domain, they share a common goal: to improve the reliability of intelligent systems by quantifying and minimizing uncertainty in complex, data-driven environments. Our study builds on this broader movement by introducing an uncertainty-aware causal framework that combines inferential rigor with the practical needs of expert decision-support systems.

Despite these advancements, most research continues to focus predominantly on improving the point-estimate accuracy of causal effect estimates, often neglecting and at the expense of addressing the critical issue of the estimation uncertainty. This overlook stems partly from the inherent complexity of simultaneously optimizing for both accuracy and inferential precision, coupled with a lack of standardized metrics and computationally efficient methods for uncertainty quantification in non-parametric causal models. However, this oversight can lead to suboptimal and potentially high-risk decisions, particularly in critical contexts such as medical treatment allocation, strategic business planning, and public policy design (Lyons & Davidson, 2016; Malekpour et al., 2020).

The increasing deployment of ML-driven causal models in high-stakes, real-world scenarios makes robust uncertainty quantification not just a theoretical endeavor, but a practical imperative for responsible decision-making. Studies such as Jesson et al. (2020) emphasize the critical role uncertainty plays in causal inference, highlighting two particularly relevant gaps for our study. First, current deep learning approaches to causal inference typically lack effective methods for identifying scenarios of “no-overlap” or violations of the positivity assumption, where the model attempts to estimate treatment effects for individuals who have little or no representation in the training data. Second, Jesson et al. point out the inadequacy of standard methods to handle covariate shifts, situations where the distribution of covariates in the training set largely differs from that of the test set or deployment scenario. This deficiency can result in overly confident and potentially misleading causal estimations. These identified gaps underscore the need for methodologies, such as those explored in our study, which not only refine the precision of causal estimates but also explicitly model and minimize estimation uncertainty to enhance robustness under conditions of non-overlap and distributional shifts.

This paper introduces an Uncertainty-Aware Causal Forest framework that treats the reliability of CATE estimates as a primary design objective. The core idea is to train causal forests to minimize finite-sample estimation uncertainty-quantified by infinitesimal-jackknife-based confidence interval (CI) width-alongside prediction error, so that the learned model explicitly accounts for accuracy and uncertainty. To achieve this, we construct a joint uncertainty-accuracy learning formulation and integrate it with Gaussian-process Bayesian optimization, which learns a probabilistic surrogate over predictive error and CI width and adaptively explores the hyperparameter space. A fully nested, three-way honest cross-validation scheme with disjoint training, validation, and test folds prevents information leakage, stabilizes uncertainty assessment, and yields credible evaluation on held-out validation and test splits of both point estimates and confidence intervals. The specific contributions are as follows:

- We introduce an Uncertainty-Aware Causal Forest learning objective that incorporates a finite-sample estimation-uncertainty term, operationalized as CI width estimated via the infinitesimal jackknife (IJ), directly into the criterion used to select splits, grow trees, aggregate leaves, and tune model complexity. This formulation leverages interval width beyond a passive, post-hoc diagnostic to an explicit quantity to be minimized during training, aligning structure discovery, regularization, and aggregation. This approach facilitates the goal of producing treatment effect estimates that are reliable at both individual and subgroup levels.
- We develop a composite hyperparameter optimization scheme based on Gaussian-process Bayesian optimization that treats predictive accuracy and estimation uncertainty as concurrent, non-collapsed objectives. This bi-objective formulation explores the hyperparameter space under an explicit trade-off between error and CI width, constructs a Pareto set of candidate configurations that span different accuracy-uncertainty preferences, and provides a principled mechanism for selecting models that are simultaneously well-calibrated, robust to overfitting, and suitable for downstream decision-making in high-dimensional, flexible settings.
- We introduce a strictly honest, three-way nested cross-validation architecture for causal forests in which tree structure learning, nuisance function estimation, hyperparameter tuning, and final evaluation are confined to disjoint data splits. This architecture couples internal honesty within trees with external training, validation, and testing folds, prevents information leakage across stages, and establishes a transparent basis for diagnosing and controlling overfitting through parallel monitoring of accuracy-oriented and uncertainty-oriented metrics on held-out data.
- We develop decision-oriented targeting policies that convert uncertainty-aware treatment effect estimates into implementable allocation rules, including a dual-threshold rule that jointly screens units on effect magnitude and confidence interval width, and a ranking rule that orders units by lower confidence bounds for use under budgetary, capacity, or fairness constraints. These policies embed statistical uncertainty directly into the intervention design step, enabling transparent, risk-aware prioritization strategies in domains such as healthcare, marketing, and infrastructure or policy planning.

Collectively, these contributions enhance the transparency and credibility of causal forests, yielding substantial reductions in estimation uncertainty while maintaining competitive predictive accuracy in benchmarks against generalized random forests (GRF) and double machine learning (DML) on one synthetic and two real-world datasets.

The remainder of this paper is organized as follows. Section 2 provides a review of the related work and essential background knowledge. Section 3 details the methodology of our proposed approach. Section 4 outlines the experimental design and discusses the results and performance evaluations of our model. Finally, Section 5 explores the implications and limitations of our findings and offers conclusions.

2. Background: Causal inference, causal forests, and uncertainty quantification

In this section, we first review key causal machine learning methods, then introduce the causal forest model and the quasi-oracle estimation of the CATE, and finally outline foundational concepts in uncertainty quantification for causal inference.

2.1. CATE estimation using machine learning algorithms

In the field of machine learning, we consistently aim to predict an outcome of interest Y from covariates X. In causal inference, our focus shifts to estimating the effect of a specific treatment, denoted by W, on Y. Here, W is a binary variable where 1 signifies receiving treatment and 0 indicates the absence of treatment. Observations with $W = 0$ are treated as control samples, whereas those with $W = 1$ are considered treated. Ideally, we seek to estimate the Individual Treatment Effect (ITE), $Y(1) - Y(0)$, for each subject. However, it is impossible to observe both potential outcomes for a single sample concurrently. This limitation corresponds to the fact that the result Y observed in the data corresponds to the potential outcome under the actual treatment assignment received, namely $Y = Y(1)$ for those who receive treatment $W = 1$ and $Y = Y(0)$ for those who do not $W = 0$. For treatment recipients, $Y(0)$ cannot be observed in the data, while for subjects without a treatment, $Y(1)$ remains unknown, in both cases, leaving the counterfactual unobserved. This fundamental challenge, often termed the "fundamental problem of causal inference" (Holland, 1986), necessitates the use of statistical and machine learning methods to estimate unobservable counterfactuals.

Since causal effects are unidentifiable at the individual level, we may look at the problem from a more general perspective. Another causal parameter that is of critical interest is the Average Treatment Effect (ATE). It represents the average effect of a treatment W on the outcome Y for all the samples. The ATE is denoted as the average difference between the two types of outcomes $Y(1)$ and $Y(0)$:

$$\Delta = \mathbb{E}[Y(1) - Y(0)] \tag{1}$$

where $\mathbb{E}[\cdot]$ denotes the expectation over the population.

However, samples have different levels of covariates and cannot share the same effect as a response to one treatment, which is called "one size may not fit all" (Guelman et al., 2015). Individuals with similar covariates may exhibit similar treatment effects. Hence, heterogeneity across covariates motivates estimation of the Conditional Average Treatment Effect (CATE), τ:

$$\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x] \tag{2}$$

where $Y(1)$ is the outcome of the samples that receive the treatment, and $Y(0)$ is the outcome of the subjects that do not receive the treatment.

Advances in machine learning have provided profound opportunities for the construction of predictive models within the realm of causal inference. A notable strategy involves integrating multiple machine learning models to assess the CATE. Established techniques such as the S-learner and T-learner employ either combined or individual model frameworks to evaluate outcomes under both treatment and control conditions (Künzel et al., 2019). Additionally, recent research has explored the application of deep learning to CATE estimation. These neural networks excel in capturing complex nonlinear relationships and interactions in high-dimensional problems (Shalit et al., 2017). Another innovative method is the double machine learning (DML) framework, introduced by Chernozhukov et al. (2018), which employs machine learning in a two-stage process to address confounding and improve upon traditional regression adjustment methods. This approach has become widely adopted through implementations such as the econml package in Python (Research, 2019), which provides flexible estimators for CATE based on the DML framework. Further advancements are highlighted in Nie and Wager (2021), where a general class of objective functions, along with a two-step algorithmic framework, is introduced. This framework first estimates marginal effects and treatment propensities, which are subsequently utilized to construct an objective function tailored to causal inference. The function is then optimized through various loss-minimization strategies, including penalized regression and deep neural networks.

2.2. Causal forest model

2.2.1. Model framework

Tree-based models, including decision trees, random forests, and gradient boosting machines, serve as foundational elements in modern machine learning, known for their interpretability and versatility in various applications for both classification and regression tasks. These models are utilized in Künzel et al. (2019) to develop metalearners that estimate heterogeneous treatment effects. The S-learner, for instance, integrates the treatment indicator as a standard feature, using it alongside other covariates to predict outcomes and calculate the CATE by altering the treatment indicator while keeping other features constant. However, this model may overlook the treatment variable if it is not a significant determinant of the outcome. Alternatively, the T-learner models treatment and control groups separately to compute CATE, but can struggle with data imbalance, potentially compromising the reliability of causal inference. To address these challenges, Künzel et al. (2019) further proposed the X-learner, which adopts a two-stage approach where separate models for treated and control groups inform the estimation of treatment effects, enhancing accuracy but increasing complexity in implementation and necessitating careful model calibration to avoid overfitting. While effective, these meta-learners generally optimize predictive loss and do not enforce the honesty constraints or uncertainty control.

Building on tree-based methods for causal inference, Wager and Athey (2018) advanced the field by introducing the causal forest, an adaptation of Breiman's random forest (Breiman, 1996) designed specifically for causal analysis. Causal forests are grounded in the potential outcomes framework, which conceptualizes causal effects as the difference between two potential outcomes, one observed under treatment ($Y(1)$) and the other under control conditions ($Y(0)$). Because only one potential outcome is observable per individual, this framework clarifies the definition and estimation of causal effects. Furthermore, causal forests rely on the assumption of unconfoundedness:

$$(Y_i(0), Y_i(1)) \perp\!\!\!\perp W_i \mid X_i \tag{3}$$

which states that, conditional on the observed covariates X_i, treatment assignment W_i is independent of the potential outcomes. Intuitively, this means that treatment decisions depend only on observable pre-treatment characteristics rather than on the unobserved outcomes themselves-an assumption that is reasonable in settings where the realized outcome is unknown at the time of assignment. Under this assumption, the CATE is point-identified and can be consistently estimated using causal-forest-based methods. In our framework, potential outcomes are estimated through orthogonalized score construction that combines separately trained outcome and propensity models, coupled with honest and nested cross-fitting to prevent information leakage and improve generalization. If treatment assignment were to depend on unobserved potential outcomes, this assumption would be violated and causal effects would no longer be point-identified without additional identifying structure such as instrumental variables, front-door adjustment, or partial-identification strategies.

Tree growth within causal forests employs recursive partitioning, guided by the "honesty" principle proposed by Athey and Imbens (2016). This principle dictates that the data used to determine the tree's splitting structure must be kept separate from the data used to estimate the treatment effects within the resulting leaves. This "honest splitting" strategy is crucial because it mitigates bias that arises from using the same data for both structural learning and parameter estimation, a common issue in traditional machine learning that can lead to overly

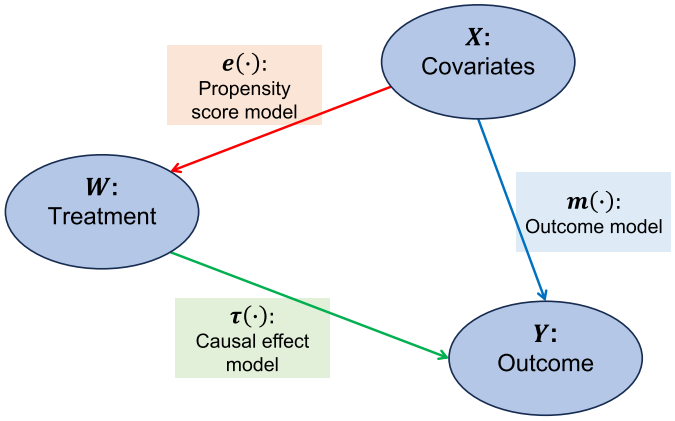


Fig. 1. Relationship between covariates, treatment, and outcome in the causal forest framework.

optimistic performance and invalid inference, especially for treatment effects.

In the causal forest framework shown in Fig. 1, two primary models are typically used to capture the relationship between covariates, treatment and outcome. The outcome model predicts the potential outcomes Y based on covariates X:

$$m(x) = \mathbb{E}[\, Y \mid X = x \,] \tag{4}$$

The idea is to model the outcome variable Y as a function of the covariates X, ignoring the treatment assignment W. This model helps in understanding how the covariates influence the outcome directly, which is crucial for adjusting the outcome predictions based on the characteristics of the individuals or units under study.

The propensity score model estimates the probability of treatment assignment given covariates:

$$e(x) = \Pr(W = 1 \mid X = x) \tag{5}$$

Modeling the treatment assignment helps in balancing the data to account for differences in the distribution of covariates among treated and control groups, thereby reducing bias in the estimation of treatment effects.

Subsequently, the causal forest estimates treatment effects by explicitly maximizing the heterogeneity in these effects across subgroups defined by the tree structures. Each split within the forest is chosen to maximize differences in estimated treatment effects between child nodes while maintaining honesty in estimation-that is, using disjoint data subsets for tree construction and effect estimation. This recursive partitioning enables the model to automatically discover regions of the covariate space where treatment responses differ most, effectively identifying subpopulations characterized by distinct combinations of features. Maximizing treatment effect heterogeneity amplifies the contrast in CATEs between distinct subgroups, allowing the model to more clearly distinguish areas of significant differential impact. By purposefully selecting splits that highlight pronounced differences in local treatment effects, the causal forest identifies finer-grained subpopulations where the treatment yields notably divergent outcomes. This targeted partitioning not only enhances interpretability but also improves the precision of treatment effect estimation. Consequently, by maximizing heterogeneity, the causal forest ensures robust and nuanced insights into how treatments operate differently across diverse population segments, thereby guiding more informed and tailored decision-making. Athey and Wager formalized this approach through the generalized random forest (GRF) methodology, and the associated grf package in R provides a widely used implementation for applied causal inference (ATHEY et al., 2019; Tibshirani et al., 2024).

2.2.2. Confidence intervals for causal forest estimates

A causal forest adapts Breiman's random forest algorithm (Breiman, 1996) to estimate heterogeneous treatment effects. Each tree partitions the covariate space and reports a leaf-specific estimate of the CATE. An ensemble of such trees is then averaged to obtain the forest predictor $\hat{\mu}(x)$. To quantify the sampling uncertainty of this predictor, Wager and Athey (Wager & Athey, 2018; Wager et al., 2014) propose the infinitesimal jackknife (IJ), which estimates the prediction variance that arises from training the forest on many resampled datasets.

For a feature vector x, the IJ variance estimator is

$$\hat{V}_{IJ}(x) = \frac{n-1}{n}\left(\frac{n}{n-s}\right)^2 \sum_{i=1}^{n} \mathrm{cov}_*\left[\hat{\mu}_b^*(x), N_{ib}^*\right]^2 \tag{6}$$

where n is the number of training samples, and s is the size of the (without-replacement) subsample used to grow each tree. $\hat{\mu}_b^*(x)$ is the single-tree estimate of the CATE at x obtained from tree b built on a subsample; the asterisk indicates the randomness arising from subsampling. N_{ib}^* is the count of how many times observation i appears in the subsample that produced tree b (0 or 1 when subsampling without replacement). $\mathrm{cov}_*\left[\hat{\mu}_b^*(x),\ N_{ib}^*\right]$ is the covariance, taken over the resampling distribution, between the tree-specific prediction at x and the inclusion count of observation i. Squaring this covariance and summing over i aggregates each observation's influence on the ensemble prediction. The term $\frac{n-1}{n}\left(\frac{n}{n-s}\right)^2$ is included as a correction for subsampling without replacement (Wager & Athey, 2018).

To establish the asymptotic normality of the causal-forest estimator $\hat{\mu}(x)$, Wager and Athey (2018) prove a central limit theorem that holds under the following regularity conditions:

(1) Honesty: each tree is built using an honest splitting rule, meaning one subsample is used to determine the tree structure (i.e., the splits), and a disjoint subsample is used to estimate treatment effects within the resulting leaves;

(2) Subsampling rate: the number of observations s used to grow each tree must increase with the total sample size n according to

$$s \asymp n^{\beta}, \qquad \beta_{\min} < \beta < 1, \tag{7}$$

with $\beta_{\min} \approx 0.5$ in practice, so that β typically lies in the interval $[0.5, 0.8]$. The notation $s \asymp n^{\beta}$ means that s grows at the same polynomial rate as n^{β} (there exist constants $c_1, c_2 > 0$ such that $c_1 n^{\beta} \le s \le c_2 n^{\beta}$ for large n). This condition balances bias and variance: as n increases, tree leaves contain more data and shrink in diameter (reducing bias), yet each tree still sees only a fraction of the sample, limiting the correlation between trees and keeping the IJ variance estimator reliable. Values of β below $\beta_{\min}$ leave leaves too coarse, while β close to 1 makes trees almost identical.

(3) Regularity conditions: standard assumptions for causal inference hold, including overlap (i.e., bounded propensity scores) and smoothness of potential outcome functions (e.g., Lipschitz continuity). Under these conditions, the estimator $\hat{\mu}(x)$ satisfies

$$\frac{\hat{\mu}(x) - \mu(x)}{\sqrt{\mathrm{Var}[\hat{\mu}(x)]}} \xrightarrow{d} \mathcal{N}(0,1), \tag{8}$$

$$\hat{V}_{\mathrm{IJ}}(x) \xrightarrow{p} \mathrm{Var}[\hat{\mu}(x)], \tag{9}$$

meaning the distribution of the normalized estimator converges in distribution to a standard normal, and the IJ variance estimator is consistent.

Because the IJ variance estimator $\hat{V}_{\mathrm{IJ}}(x)$ is consistent, an asymptotically valid two-sided $100(1-\alpha)\%$ confidence interval (CI) for the CATE at x is

$$\mathrm{CI}_{100(1-\alpha)\%} = \hat{\mu}(x) \pm z_{1-\alpha/2}\sqrt{\hat{V}_{\mathrm{IJ}}(x)} \tag{10}$$

with $z_{1-\alpha/2}$ the $(1-\alpha/2)$-quantile of the standard normal distribution (e.g. $z_{0.975} = 1.96$ for a 95 % interval).

Honest splitting eliminates bias from using the same data to choose partitions and estimate treatment effects, while the $s \asymp n^{\beta}$ rule guarantees that leaf diameters contract as more data arrive yet individual trees remain weakly dependent. Together these conditions yield both a

Gaussian limit for $\hat{\mu}(x)$ and a consistent IJ variance estimator, enabling principled uncertainty quantification for causal-forest CATE estimates.

2.3. Quasi-oracle estimation of the CATE

As discussed in Section 2.1, a fundamental challenge in causal inference is that the outcomes for both receiving treatment and not receiving treatment cannot be observed simultaneously for the same individual. This implies that the true causal effect, $Y(1) - Y(0)$, is never directly observable, even in training data.

To address this issue, a quasi-oracle estimation approach was proposed in Nie and Wager (2021). This method consists of two main steps: In stage 1, an approximation of the outcome model $\hat{m}$ and propensity score model $\hat{e}$ is learned for the oracle objective. Specifically, $\hat{m}$ is estimated by fitting covariates X to outcomes Y, while $\hat{e}$ is learned by fitting covariates X to treatment indicators W. In stage 2, the learned approximation is optimized by minimizing a squared loss function. The treatment effect $\hat{\tau}$ with the minimum squared loss is found in this stage. This two-stage procedure is designed to enable the construction of an objective function for causal effect estimation that effectively mimics an oracle.

Specifically, the quasi-oracle objective function for the treatment effect $\tau(\cdot)$ is defined as:

$$\hat{L}_n\{\tau(\cdot)\} = \frac{1}{n}\sum_{i=1}^{n}\left[\left(Y_i - \hat{m}^{(-q(i))}(X_i)\right) - \left(W_i - \hat{e}^{(-q(i))}(X_i)\right)\tau(X_i)\right]^2 \tag{11}$$

The CATE function $\hat{\tau}(\cdot)$ is then estimated by solving:

$$\hat{\tau}(\cdot) = \arg\min_{\tau}\left[\hat{L}_n\{\tau(\cdot)\} + \Lambda_n\{\tau(\cdot)\}\right] \tag{12}$$

In this setting, the dataset is partitioned into Q equally sized folds following a quasi-oracle estimation strategy. For each fold q, the data are split such that the fold itself is used for estimating τ, while the remaining folds are used to train the nuisance models $\hat{m}$ and $\hat{e}$. The superscript $(-q(i))$ indicates that predictions for the ith observation are obtained out-of-bag, i.e., without using data from the fold to which observation i belongs. Formally, X_i, W_i, and Y_i denote the covariates, treatment assignment, and observed outcome, respectively. The terms $\hat{m}^{(-q(i))}(X_i)$ and $\hat{e}^{(-q(i))}(X_i)$ represent the predicted outcome and propensity score for unit i, trained without using its own fold. Eq. (12) then seeks the treatment effect function $\hat{\tau}$ that minimizes the empirical loss $\hat{L}_n\{\tau(\cdot)\}$-the squared discrepancy between observed and predicted outcomes-while incorporating a regularization penalty $\Lambda_n\{\tau(\cdot)\}$ that controls complexity of the function class (e.g., penalized regression or a nonparametric learner) to mitigate overfitting.

2.4. Uncertainty in causal inference

Uncertainty presents a significant challenge in causal inference, particularly when it affects decision-making based on estimated causal effects. This issue becomes particularly pronounced when considering whether to administer a treatment to a new subject whose covariate profiles may be underrepresented in the training data. Reliable causal conclusions, therefore, depend not only on accurate point estimates but also on a transparent quantification of their uncertainty.

In general, the uncertainty of causal inference arises from three main sources (Abdar et al., 2021; Jesson et al., 2020; Van Amersfoort et al., 2021). First, sampling variability reflects finite-sample noise and random variation inherent in the observed data. Second, model uncertainty stems from the specification and estimation process, including the stochasticity of machine-learning models and potential misspecification of the nuisance functions (outcome and propensity score models). Third, identification uncertainty arises when the core assumptions of causal inference-particularly unconfoundedness and positivity-are partially violated. These three sources interact to influence both the bias and variance of estimated treatment effects.

A pivotal element contributing to this uncertainty is the violation of the positivity assumption in causal analysis. The positivity assumption stipulates that for every subject, the conditional probability of receiving the treatment, given their covariates, must be greater than zero but less than one. Mathematically, this is expressed as:

$$0 < P(W_i = 1 \mid X_i = x) = e(x) < 1, \quad \forall x \in \text{support of } X_i \tag{13}$$

This requirement is crucial because it ensures that each individual has a non-zero probability of receiving and not receiving the treatment, allowing for a fair comparison across different covariate profiles. Violations of this assumption often occur in practice when certain subgroups within the data rarely, if ever, receive particular treatments, leading to covariate spaces where the assumption fails. Such scenarios result in a lack of treatment diversity, thereby significantly undermining the reliability of causal effect estimates and introducing substantial uncertainty into the analytical process.

Another important contributor is covariate shift, the difference between the covariate distributions of the training and test data. Such discrepancies arise when the conditions prevalent during model training do not align with those in which the model is subsequently applied or evaluated. This mismatch can substantially degrade model performance because the predictive model may not accurately capture or adjust to new data environments. As a result, the reliability of the model's predictions, crucial for informed decision-making in causal inference, is compromised, introducing significant uncertainty into the analysis.

Building on these insights, the literature reveals a focused exploration of uncertainty within the treatment and control groups using methods like paired and matched set bootstrapping (Li & Cho, 2023). Although these methods provide insights into the distinct uncertainties characterizing each group, a significant gap persists in quantifying how these uncertainties impact the overall accuracy of causal effect estimations in non-randomized settings. This underscores the need for advanced methodologies that can measure and mitigate these uncertainties to bolster the dependability of causal inference analyses.

3. Methodology

In this section, we first present our proposed methodological framework, summarized in Fig. 2. The framework consists of three key components: a structured nested *honest* cross-validation approach to mitigate overfitting and prevent information leakage, a two-stage optimization strategy to jointly minimize estimation uncertainty – operationalized as CI width via the infinitesimal jackknife – and mean squared error (MSE), and the integration of machine learning models within the causal forest framework with hyperparameter tuning performed via Gaussian process-based Bayesian optimization that co-optimizes prediction error and CI width. Finally, the framework introduces a dual-threshold policy targeting rule that leverages both estimated CATEs and their confidence intervals to guide practical treatment allocation decisions.

3.1. Proposed cross-validation process to address overfitting in CATE estimation

Overfitting poses a critical challenge in causal forest-based CATE estimation, where excessive model flexibility can lead to overly optimistic in-sample performance but poor generalization and inflated confidence in treatment decisions. From a machine-learning standpoint, deep trees and large ensembles reduce bias but amplify variance and uncertainty if unchecked. From a causal-inference perspective, failing to rigorously separate data used for (i) tree construction, (ii) nuisance estimation, and (iii) model selection/evaluation can introduce "information leakage", invalidate asymptotic variance approximations, and undermine the honesty guarantees central to unbiased CATE estimation. In addition, leakage between folds can bias IJ variance estimates downward, yielding over-narrow confidence intervals.

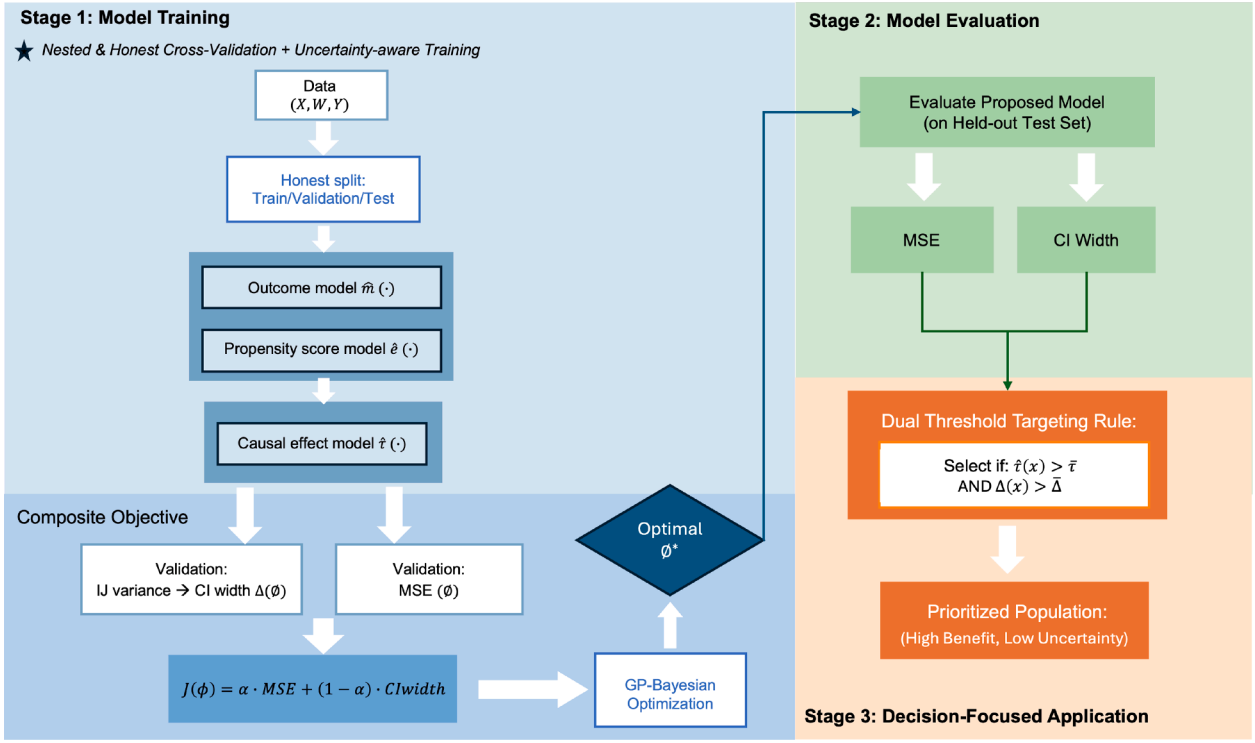


Fig. 2. Overview of the proposed three-stage Uncertainty-Aware Causal Forest framework. Stage 1 (Model Training) employs nested and honest cross-validation together with a bi-objective validation criterion-mean squared error (MSE) and infinitesimal-jackknife (IJ) confidence interval width-optimized via Gaussian-process Bayesian optimization. Stage 2 (Model Evaluation) assesses the selected model on a held-out test set using both predictive accuracy (MSE) and uncertainty (CI width). Stage 3 (Decision-Focused Application) applies a dual-threshold, risk-aware targeting rule that prioritizes individuals with high estimated treatment effects and low uncertainty, linking robust causal estimation to transparent decision making.

Fig. 3 presents a comparison of training procedures and objective functions across three machine learning models for causal effect estimation, including the proposed method. In honest causal forest (Wager & Athey, 2018), denoted as Model (1), the bootstrap sample set s is divided into two equally sized subsets, with one half building the partition model, and the other used to estimate treatment effects within the resulting partitions. This method, known as "honest" splitting, is designed to prevent overfitting by ensuring that the data used to construct the model structure is separate from the data used for effect estimation. While this reduces within-leaf variance and bias, it does not incorporate a separate validation set for fine-tuning parameters or structures during training, which can lead to challenges in controlling model complexity and may result in either underfitting or overfitting due to suboptimal model complexity decisions. Moreover, hyperparameters selected on the same data used to build trees can implicitly couple structure and tuning, again risking optimistic inference.

The quasi-oracle method (Nie & Wager, 2021), referred to as Model (2) here, adopts a cross-validation approach where one fold of data is reserved exclusively to estimate the causal effect, while the remaining folds are used to develop the outcome and propensity score models. This structure aims to balance the need for accurate effect estimation with the robustness of the model against variations in data. However, it is important to note that the quasi-oracle advantage of this method presupposes that initial estimates of marginal effects and treatment propensities are fairly accurate. Significant deviations in these preliminary estimates can adversely affect the model's efficacy. Further, Model (2) typically does not include a distinct validation fold dedicated to hyperparameter selection for the final causal learner, leaving model complexity partially uncontrolled and creating opportunities for fold reuse.

Unlike the quasi-oracle method of Nie and Wager [29], which reserves only one fold for treatment effect estimation but reuses the same data for model selection, our nested structure (Model (3) in Fig. 3) introduces a distinct validation set for hyperparameter tuning. This additional layer eliminates dependence between model fitting and parameter selection, thereby mitigating both overfitting and information leakage. This approach systematically prevents overfitting while jointly optimizing predictive accuracy and uncertainty reduction. Specifically, we use a training set to simultaneously estimate the outcome, propensity score, and treatment effect models, a validation set to tune parameters and select optimal model specifications, and a testing set to provide an unbiased evaluation of the final model. The introduction of a distinct validation set allows for effective parameter adjustment and model optimization and ensures that tuning decisions reflect generalization beyond the training sample without risking the integrity of the test data. This strategy promotes an ideal balance between bias and variance, preparing the model for its final evaluation. A separate testing set is used for this final assessment, ensuring that the model's performance is measured against completely unseen data, providing a reliable indicator of its real-world efficacy. Practically, (i) we stratify all folds by treatment to preserve overlap; (ii) we perform cross-fitting within the training fold when learning $\hat{m}$ and $\hat{e}$ to avoid in-fold bias; (iii) we evaluate the outer objective on the validation fold – CI width via IJ as the primary target, with MSE monitored for accuracy– and pass these fold-wise scores to a Gaussian-process Bayesian optimizer; and (iv) we lock hyperparameters before final refit and compute all test metrics (including IJ CIs) on data never used for training or tuning.

Analytically, our three-way split enforces honest estimation at every stage by allocating disjoint data subsets for tree construction, nuisance-model fitting, hyperparameter tuning, and final evaluation. This strict separation eliminates any correlation between the samples used to learn split structure and the samples used to estimate leaf-level effects or to select hyperparameters, preserving the unbiasedness of each estimator and satisfying the "honesty" conditions required for valid variance approximation in causal forests. Moreover, this approach stabilizes the

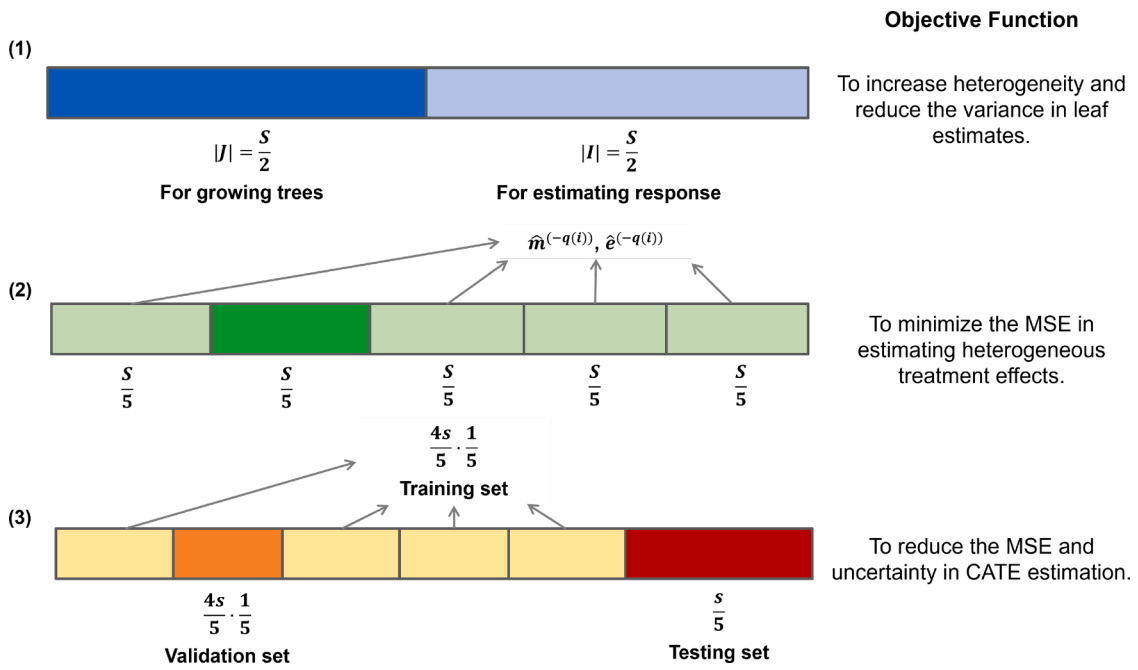


Fig. 3. Comparison of training procedures and objective functions in existing and proposed models. (1) the honest model in Wager and Athey (2018). (2) the two-step quasi-oracle model in Nie and Wager (2021). (3) our model.

Gaussian-process surrogate in Bayesian optimization by providing noise-reduced, independent evaluations of our composite objective–including MSE and CI width–on a dedicated validation fold. Because these measurements are not contaminated by the same data used to train the trees or fit nuisance functions, the surrogate's posterior variance shrinks more rapidly, improving acquisition decisions and accelerating convergence to the true hyperparameter region that minimizes CI width subject to accuracy.

Finally, by reserving a completely independent test set, we ensure that the IJ variance estimates are computed on data that never influenced either tree-structure learning or hyperparameter tuning. In classical asymptotic theory for ensemble methods (Wager & Athey, 2018), unbiased, independent contributions from each subsampled tree are required to invoke a central-limit argument: when the number of trees grows and each tree's prediction uses fresh data, their average converges to a normal distribution. Our three-way split enforces this "freshness" by guaranteeing that the test set plays no role in model fitting or parameter selection, so each tree's effect estimate on that set is statistically independent of the training procedure. Under these conditions, the IJ estimator, which is designed to consistently approximate the true sampling variance of a bagged predictor, converges to the correct variance as sample size increases. Consequently, the derived CIs attain their nominal coverage in large samples, even in high-dimensional settings, because (i) the honest forest structure yields near-independent tree contributions, (ii) the IJ captures their aggregate variability, and (iii) the withheld test data prevent any bias from tuning or overfitting from contaminating the variance estimate. Empirically, this protocol also reduces the train-test gap for both CI width and MSE, indicating improved generalization.

In high-dimensional causal inference, reliable estimation is challenged by three interrelated factors: (i) overfitting, where flexible learners capture noise rather than true treatment heterogeneity; (ii) data leakage, where reusing the same samples for tree construction, nuisance estimation, and model tuning leads to biased effect estimates and overly narrow confidence intervals; and (iii) model misspecification, where unstable or poorly calibrated nuisance models propagate bias to CATE estimates. The proposed three-way nested validation framework systematically mitigates these issues by enforcing strict data separation among training, validation, and testing folds, ensuring that model fitting, hyperparameter optimization, and performance evaluation remain independent. Within each training fold, cross-fitting stabilizes the nuisance outcome and propensity models even in high-dimensional settings, while the outer validation layer jointly optimizes predictive accuracy and confidence interval width to balance bias and variance. Together, these components prevent overfitting and information leakage, yielding uncertainty estimates that reflect genuine epistemic variability rather than distortions induced by the training procedure. By enabling reproducible evaluation on held-out validation and test splits, the framework establishes a principled foundation for trustworthy causal analysis in complex, high-dimensional applications.

To make the validation process explicit, the complete three-way nested framework can be summarized as follows: (i) the data are partitioned into three independent subsets-*training*, *validation*, and *testing*; (ii) within the training subset, outcome and propensity score models are learned and cross-fitted to eliminate in-fold bias; (iii) the validation subset is used exclusively for hyperparameter selection via Gaussian-process Bayesian optimization, where both MSE and IJ-based CI width are jointly evaluated as tuning criteria; (iv) once the optimal configuration is identified, the model is refitted on the combined training and validation data, and the testing subset provides an independent, unbiased assessment of performance; and (v) the reported training, validation, and testing metrics in Section 4.3.3 correspond directly to these three stages. This clarification emphasizes that model training, tuning, and evaluation are fully disjoint, ensuring that validation outcomes reflect genuine generalization rather than overfitting or information leakage.

3.2. *Composite objectives: Accuracy and uncertainty*

Building on the uncertainty sources discussed in Section 2.4, we design a composite optimization strategy that jointly enhances the accuracy and reliability of causal forest estimation. The central idea is to decouple the objectives of predictive accuracy and uncertainty precision by structuring the training process into two distinct stages: an inner layer focused on minimizing MSE of the nuisance models, and an outer layer that explicitly minimizes the width of the estimated CATE confidence intervals. This two-stage framework allows us to preserve the

orthogonality properties necessary for unbiased treatment effect estimation, while also optimizing for inferential precision.

Formally, let $\phi \in \mathcal{S}$ denote the set of causal-forest hyperparameters. The hyperparameter set defines the structural configuration of the causal forest. Specifically, these parameters control the number of trees in the ensemble, the maximum allowable depth of each tree, the minimum number of samples required to perform an internal split, and the minimum number of observations permitted in each terminal leaf node. Together, they regulate the trade-off between model flexibility and estimation variance, ensuring that the forest is sufficiently expressive without overfitting.

For given nuisance models $\hat{m}(x)$ and $\hat{e}(x)$, the individualized treatment effect estimator $\hat{\tau}(x;\phi)$ is obtained by fitting a causal forest parameterized by ϕ. The proposed joint optimization seeks to minimize a composite objective that balances predictive accuracy and uncertainty:

$$J(\phi) = \alpha \cdot \mathrm{MSE}(\phi) + (1-\alpha)\cdot \mathrm{Var}(\phi), \tag{14}$$

where $\alpha \in [0,1]$ controls the trade-off between accuracy and uncertainty. The two components are defined as

$$\mathrm{MSE}(\phi) = \frac{1}{n_{\mathrm{val}}} \sum_{i \in D_{\mathrm{val}}} \left(Y_i - \hat{m}(X_i) - (W_i - \hat{e}(X_i))\,\hat{\tau}(X_i;\phi)\right)^2, \tag{15}$$

$$\mathrm{Var}(\phi) = \frac{1}{n_{\mathrm{val}}} \sum_{i \in D_{\mathrm{val}}} \frac{\left(u_i(\phi) - \ell_i(\phi)\right)^2}{4 \times 1.96^2}, \tag{16}$$

where $[\ell_i(\phi), u_i(\phi)]$ denotes the 95 % confidence interval for the estimated CATE at observation i. The optimal configuration of hyperparameters is obtained by

$$\phi^\star = \arg\min_{\phi \in \mathcal{S}} J(\phi), \tag{17}$$

where evaluation of $J(\phi)$ is performed via Bayesian optimization with Gaussian-process surrogates and K-fold cross-validation across $\mathcal{D}_{\mathrm{train}}$. Details of the hyperparameter space, search ranges, and justification are provided in Appendix B.

In the inner layer, the causal forest relies on doubly robust score construction, which requires two nuisance components: an outcome model $\hat{m}(x)$ and a propensity score model $\hat{e}(x)$. We estimate both functions using flexible base learners, random forests or Lasso regressors, whose parameters are implicitly optimized by minimizing MSE during each internal call to the causal forest's fitting routine. We use cross-fitting to obtain out-of-fold nuisance predictions, preventing in-fold bias and maintaining orthogonality. Following the two-stage principles described in Eqs. (11) and (12), the causal effect model $\hat{\tau}(x)$ is constructed using the empirically optimal nuisance estimates $\hat{m}(x)$ and $\hat{e}(x)$. By anchoring the orthogonalized score to these high-quality predictions, the approach reduces bias and stabilizes the variance of the resulting CATE estimates.

The outer layer targets uncertainty directly by minimizing the average squared width of the CIs around $\hat{\tau}(x)$. For each hyperparameter configuration $\phi \in \mathcal{S}$, we perform K-fold cross-validation. In each fold, the nuisance models and causal forest are trained on the corresponding training subset, and 95 % CATE CIs $[\ell_i, u_i]$ are computed on the held-out validation set via the infinitesimal jackknife. We formalize this via the loss function $\hat{L}_n$, defined as the mean squared half-width of the 95 % CIs across n samples:

$$\hat{L}_n\{\tau(\cdot)\} = \frac{1}{n}\sum_{i=1}^{n} \frac{1}{4\times(1.96)^2}\left[\text{upper bound}_{CI}(\tau(\cdot)) - \text{lower bound}_{CI}(\tau(\cdot))\right]^2 \tag{18}$$

We subsequently estimate the CATE function by solving:

$$\hat{\tau}(\cdot) = \arg\min_{\tau}\left[\hat{L}_n\{\tau(\cdot)\}\right] \tag{19}$$

The outer objective function $f(\phi)$ is then defined as the mean of these losses across all folds, providing a direct estimate of estimator variance.

The full procedure is summarized in Algorithm 1, which outlines the nested structure of MSE-driven model fitting and CI-width-based outer-loop tuning. The algorithm concludes by fitting a final causal forest model using $\phi^\star$ on the entire training set, yielding a ready-to-use estimator $\hat{\tau}(x)$ with minimized uncertainty and robust treatment effect estimation.

Algorithm 1: Two-layer hyperparameter optimization for causal forest (Inner MSE → outer CI-width).

Input: Training data $\mathcal{D}_{\mathrm{train}} = (X_{\mathrm{train}}, W_{\mathrm{train}}, Y_{\mathrm{train}})$, Hyperparameter search space $\mathcal{S}$, number of CV folds K
Output: Optimal hyperparameters $\phi^\star$ and final causal forest model $\tau(\cdot)$

Layer 1: Implicit MSE minimization
  Base learners $m(\cdot), e(\cdot) \leftarrow$ RF regressors or Lasso regressors.
  foreach $(X, Y, W) \in \mathcal{D}_{train}$ **do**
    $m_Y^\star \leftarrow \arg\min_{m_Y} \frac{1}{|D|} \sum_{(x,y,w)\in D} (y - \hat{y}(x; m_Y))^2$
    $e_W^\star \leftarrow \arg\min_{e_W} \frac{1}{|D|} \sum_{(x,y,w)\in D} (w - \hat{w}(x; e_w))^2$
  Use $m_Y^\star$, $e_W^\star$ to compute CATE estimates $\hat{\tau}(x)$.
Layer 2: Explicit CI width Minimization.
Define the loss for a validation fold

$$\mathcal{L}_{\mathrm{val}}(\phi) = \frac{1}{|\mathcal{D}_{\mathrm{val}}|}\sum_i \frac{(u_i - \ell_i)^2}{4\cdot 1.96^2},$$

where $[\ell_i, u_i]$ is the 95 % CI for $\hat{\tau}(x_i)$.
Bayesian optimisation loop.
foreach *candidate* $\phi \in \mathcal{S}$ *chosen by the optimiser* **do**
  for $k = 1$ **to** K **do**
    Split $\mathcal{D}_{\mathrm{train}} \to \left(D_k^{\mathrm{train}}, D_k^{\mathrm{val}}\right)$.
    Fit $m_Y^\star$, $e_W^\star$ on D_k^{train}.
    Compute $\mathcal{L}_k(\phi)$ on D_k^{val} using the CI-width formula.
  $f(\phi) \leftarrow \frac{1}{K}\sum_{k=1}^{K} \mathcal{L}_k(\phi)$
$\phi^\star \leftarrow \arg\min_{\phi\in\mathcal{S}} f(\phi)$.
Final fit and return.

Hyperparameter tuning is often treated as a purely technical step, yet in causal inference, it directly shapes the credibility of both point estimates and the confidence intervals used for decision-making. By explicitly optimizing for uncertainty rather than relying solely on predictive accuracy, our two-layer strategy ensures that the resulting treatment effect estimates are not only unbiased but also accompanied by well-calibrated, informative confidence intervals. This matters in high-stakes policy and business settings where the reliability of causal estimates must be as transparent as their magnitude (Rehill & Biddle, 2024). Our framework elevates hyperparameter optimization from a heuristic process to a principled component of causal learning, aligning the model's internal objectives with the inferential demands of real-world applications. Additional quantitative validation of the joint optimization framework-including convergence behavior, bias-variance trade-offs, and uncertainty calibration-is provided in Appendix C.

3.3. Outcome model and propensity score model

Reliable causal-forest inference begins with high-quality estimates for the outcome model $\hat{m}(\cdot)$ and the propensity score model $\hat{e}(\cdot)$. We obtain these functions using flexible base learners, random forests or Lasso

regressors, whose parameters are implicitly optimized by minimizing MSE during each internal call to the causal forest's fitting routine.

3.3.1. Random forest regressors

The outcome model $\hat{m}(\cdot)$ and propensity score model $\hat{e}(\cdot)$ are two separate regression forests in default causal forest models (Tibshirani et al., 2024). A random forest is an ensemble learning method based on decision tree algorithms. It operates by constructing a large ensemble of decision trees during training and outputting the average prediction of the individual trees. Random forests can handle complex datasets with nonlinear relationships and interactions between variables. It is less likely to overfit because it averages multiple deep decision trees, each trained on different parts of the same training set.

For a regression problem, the prediction of a random forest model is typically the average of the predictions from all the individual trees. Mathematically, if there are N trees in the forest and $\hat{y}_i^{(b)}$ represents the prediction of the bth tree for the ith sample, the final prediction $\hat{y}_i$ for that sample is given by:

$$\hat{y}_i = \frac{1}{N}\sum_{t=1}^{N}\hat{y}_i^{(b)} \tag{20}$$

3.3.2. Lasso regressors

In our two-stage optimization framework, the causal forest relies on precise first-stage estimates of the outcome model $\hat{m}(\cdot)$ and the propensity score model $\hat{e}(\cdot)$. Besides using the conventional random forest base learners, we also adopt the Least Absolute Shrinkage and Selection Operator (Lasso) regressors. The choice of Lasso is motivated by its superior performance in high-dimensional and approximately sparse settings commonly encountered in practice. Specifically, the Lasso incorporates an ℓ_1-penalty that effectively shrinks irrelevant coefficients to zero, automatically selecting a small subset of covariates with the greatest predictive power. This feature ensures that the first-stage estimates achieve the $n^{-1/2}$ convergence rate necessary to maintain valid orthogonal scores within the causal forest framework.

Moreover, by imposing sparsity, Lasso regressors significantly reduce the variance of the estimators, enhancing the stability and interpretability of the resulting models. This is particularly valuable for understanding and communicating the role of specific covariates in the causal estimation process. Additionally, modern implementations of Lasso leverage computationally efficient coordinate-descent algorithms, enabling rapid refits during each iteration of the Bayesian hyperparameter optimization loop. Thus, Lasso regressors not only provide theoretical guarantees in terms of accuracy and consistency but also practical advantages in terms of computational efficiency and interpretability. Collectively, these properties position Lasso as an optimal choice for first-stage learners within the causal forest modeling pipeline.

Formally, Lasso solves the penalised least-squares problem (Tibshirani, 1996)

$$\min_{\beta_0,\{\beta_j\}_{j=1}^{p}} \frac{1}{2n}\sum_{i=1}^{n}\Big(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\Big)^2 + \alpha\sum_{j=1}^{p}|\beta_j|, \tag{21}$$

where y_i is the response, x_{ij} denoted the j-th covariate of observation i, β_0 is the intercept, and $\alpha > 0$ controls the strength of the ℓ_1-penalty. Larger values of α induce greater sparsity, while setting $\alpha = 0$ reverts the optimization problem to ordinary least squares regression. Taken together, these attributes affirm Lasso regressors as a theoretically robust and computationally efficient choice for the first-stage learners within the causal forest modeling pipeline.

3.4. Hyperparameter searching

Hyperparameter selection strongly affects the bias-variance trade-off in causal forest models, influencing both predictive accuracy and uncertainty estimates. Unlike earlier work that solely optimized hyperparameters for MSE, our approach employs a composite objective that simultaneously accounts for uncertainty and predictive accuracy, providing a novel and more balanced perspective on hyperparameter tuning. To systematically navigate the high-dimensional hyperparameter space, we implement a structured optimization strategy utilizing Bayesian optimization with Gaussian processes.

3.4.1. Bayesian optimization with Gaussian processes

We employ Bayesian optimization with Gaussian process (GP) regression to efficiently identify optimal hyperparameter settings. Bayesian optimization is particularly advantageous for optimizing complex, computationally expensive objective functions, such as those encountered in causal forest model tuning (Frazier, 2018; Snoek et al., 2012; Wang et al., 2024). In our implementation, we specifically optimize the causal forest's variance of the CATE estimates as measured by the average confidence interval width derived through cross-validation. We fix cross-validation splits and random seeds so that fold-wise evaluations of CI width are comparable across candidates, and we model observation noise in the GP to account for Monte-Carlo variability. Hyperparameters are selected based on performance on the validation fold, which acts as a proxy for unseen data; this penalizes configurations that overfit the training set and guides the optimizer toward models with stronger generalization on held-out validation and test splits.

To achieve this, we define the objective function as the cross-validated average variance of the CATE estimates, calculated from the width of 95 % CIs produced by the causal forest model. At each iteration, Bayesian optimization leverages a GP surrogate model to balance the exploration of unexplored hyperparameter combinations and exploitation of regions already known to yield lower variance estimates. We maximize an acquisition function, specifically Expected Improvement, to propose new candidate hyperparameters. The hyperparameters optimized include the number of trees, maximum tree depth, minimum samples required to split an internal node, and minimum samples per leaf. These parameters regulate model complexity, directly influencing the bias-variance trade-off in the causal forest.

3.4.2. Hyperparameter smoothing

To further refine the identified optimal hyperparameter regions, we apply hyperparameter smoothing after the initial Bayesian optimization phase. This process involves empirically defining narrower hyperparameter intervals informed by the best-performing hyperparameter settings identified during Bayesian optimization. Specifically, we compute quantiles from the top-performing configurations, establishing an informed "trust region".

Within this restricted subspace, we retrain a secondary GP model using only these optimal configurations, thereby refining our posterior distribution. This targeted refinement reduces computational complexity and accelerates convergence by focusing the search on the most promising areas. Additionally, hyperparameter smoothing helps avoid local optima by promoting smoother transitions within the hyperparameter search space. This iterative process continues, updating quantile-based constraints and retraining the GP model, until incremental improvements in the acquisition function fall below a predefined threshold, signaling convergence. Finally, we lock the selected hyperparameters and refit on the union of training and validation data under honesty, then compute all metrics and IJ-based CIs on the untouched test set to ensure unbiased assessment.

3.5. Uncertainty-aware treatment allocation policy

In many decision-making contexts, such as allocating costly treatments or scarce resources, it is not sufficient to estimate individual-level treatment effects. A critical downstream task is to identify which subjects should be prioritized for intervention. To this end, we introduce a simple yet effective population targeting rule that selects individuals expected to benefit most from treatment while simultaneously accounting for estimation reliability. This approach connects heterogeneous treatment effect estimation with practical policy targeting, a linkage

emphasized in recent work on individualized decision rules and policy learning (Athey & Wager, 2017, 2021; Kitagawa & Tetenov, 2018).

Formally, let $\hat{\tau}(X_i)$ denote the estimated CATE for subject i with covariates X_i, and let $\hat{\Delta}(X_i)$ denote the width of the associated 95 % CI. Our targeting criterion selects subjects who satisfy both:

$$\hat{\tau}(X_i) > \overline{\tau}, \tag{22}$$

$$\hat{\Delta}(X_i) < \overline{\Delta}, \tag{23}$$

where $\overline{\tau}$ is the mean estimated CATE across the population and $\overline{\Delta}$ is the mean CI width. In words, we target individuals who (i) are predicted to experience above-average treatment benefits and (ii) for whom the estimates are relatively precise.

An equivalent risk-aware selection is to rank by a lower-confidence bound (LCB),

$$\mathrm{LCB}_i \;=\; \hat{\tau}(X_i) \;-\; z_{1-\alpha}\sqrt{\hat{V}_{\mathrm{IJ}}(X_i)}, \tag{24}$$

and select either the top-K units (budgeted scenario) or those exceeding a policy threshold. The $\hat{\tau}(X_i)$ denotes the estimated CATE for subject i and the variance estimator $\hat{V}_{\mathrm{IJ}}(X_i)$ is defined in Eq. (6).

The dual-threshold rule above serves as an interpretable surrogate for this LCB ranking. When costs vary across units, one can solve a budgeted assignment (knapsack) problem:

$$\max_{a_i\in\{0,1\}} \quad \sum_i \mathrm{LCB}_i\, a_i \tag{25}$$

$$\text{s.t.} \quad \sum_i c_i\, a_i \;\le\; B, \tag{26}$$

where a_i indicates assignment, c_i is unit cost, and B is the total budget. Additional operational or fairness constraints (e.g., capacity or priority groups) can be incorporated linearly without altering the risk-aware objective.

This dual-threshold approach balances effect heterogeneity with estimation certainty. By avoiding reliance on extreme but highly uncertain estimates, the rule reduces the risk of allocating treatment to individuals whose apparent benefits are driven by noise. In practice, this ensures that scarce resources are directed to individuals with both high expected impact and robust statistical support, making the targeting strategy particularly well suited for decision-support applications (Nie & Wager, 2021).

4. Experiments and results

In this section, we evaluate the proposed framework using both synthetic and real-world datasets. We begin by introducing the datasets, then describe the experimental setup, including model configurations and evaluation metrics. Finally, we present and analyze the results, comparing our approach against established benchmarks.

4.1. Data

We begin by detailing the process of generating the synthetic data, followed by a description of two empirical datasets.

Job Training (Synthetic data): We generated a synthetic dataset of 10,000 samples, following a structured approach inspired by similar methodologies (Nie & Wager, 2021). The data generation process involved the following steps:

$$X_i \sim U_d(0,1) \tag{27}$$

$$W_i \sim \text{Bernoulli}(0.5) \tag{28}$$

$$\epsilon_i \sim N(0,1) \tag{29}$$

$$Y_i = m^*(X_i) + W_i\tau^*(X_i) + \sigma\epsilon_i \tag{30}$$

For each subject i, the feature set X_i is drawn from a uniform distribution across d dimensions. We define $d = 2$ to correspond to the variables of education level $X_{1,i}$ and age $X_{2,i}$ for each individual. The treatment variable W_i is modeled as a Bernoulli distributed variable with a probability of 0.5. Additionally, the error term ϵ_i is sourced from a standard normal distribution, where the standard deviation σ is set to 1. The baseline main effect and the treatment effect function are set as $m^*(X_i) = 3000 + 100X_{1,i} + 5X_{2,i}$ and $\tau^*(X_i) = 200 + X_{1,i} - 0.5X_{2,i}$, respectively. This construction yields heterogeneous treatment effects with a simple, interpretable linear structure in X and ensures overlap by design.

National Study of Learning Mindsets (Mindsets): The National Study of Learning Mindsets, a randomized trial carried out in U.S. public high schools, aimed to assess the effectiveness of a nudge-like intervention aimed at fostering a growth mindset in students and its impact on academic achievement. To protect privacy, we use the publicly released, model-based simulated dataset calibrated to the original study (as in the 2018 Atlantic Causal Inference Conference workshop) (Yeager et al., 2016). The dataset comprises 10,391 samples and can be accessed as described in Athey and Wager (2019).

Infant Health and Development Program (IHDP): The IHDP is a randomized controlled trial designed to evaluate the impact of specialist home visits on children's cognitive test scores. The dataset contains real covariates and treatment assignments, but the outcomes are semi-synthetic, simulated conditional on the observed covariates and treatments. To ensure a clear separation between treatment and control groups, units from non-white mothers in the treatment arm are excluded, as described in Hill (2011). Following the standard protocol, we adopt the 672/75 train-test split and exploit all 100 semi-synthetic replications. For each replication, models are trained on the 672 training units and evaluated on the corresponding 75 test units, with final performance reported as the mean and dispersion across the 100 runs. Additional exploratory visualizations and balance diagnostics for all datasets are provided in Appendix A.

4.2. Experiment design

In our experiment, the design focuses on achieving two primary objectives: MSE minimization and uncertainty reduction in the context of causal inference. This experimental setup involves a comparison between our proposed method and the generalized random forest (GRF) method and the double machine learning (DML) framework. Each of the datasets is randomly divided into an 80 % training set and a 20 % test set. Additionally, 20 % of the training set (equivalent to 16 % of the total dataset) is designated as the validation set. This partitioning facilitates both model training and robust validation of predictive accuracy and causal effect estimation.

4.3. Experimental results

4.3.1. Results of CATE estimation

Fig. 4 presents the smoothed CATE estimates and their 95 % confidence intervals on the synthetic dataset, comparing results from the baseline DML approach with those obtained after training using the proposed method. The estimates are plotted against an observation index sorted by education level. To reduce local fluctuations and highlight systematic patterns, both the CATE estimates and their confidence intervals were smoothed using a moving average with a window size of 30. While smoothing is not strictly required in causal inference, it is commonly applied in practice to make heterogeneous treatment effect patterns more interpretable (Athey & Imbens, 2016; Wager & Athey, 2018). In this context, we use the term *stable* to indicate that the estimated CATE curve exhibits fewer erratic fluctuations across adjacent observations, reflecting lower local variability. Such stability is desirable because it suggests that the model is capturing persistent heterogeneity patterns driven by

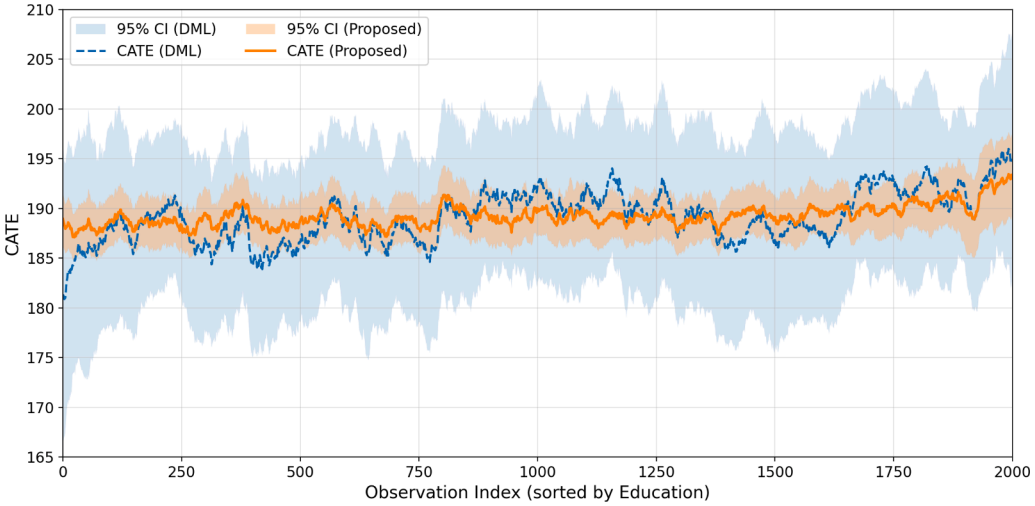


Fig. 4. Comparison of smoothed CATE estimates on a 2,000-observation subsample of the 10,000-point synthetic data (DML vs. proposed method).

covariates (e.g., education level) rather than noise, thereby yielding narrower confidence intervals and more precise treatment effect estimation. This reduction in estimation uncertainty highlights the effectiveness of the training procedure in improving both the precision and reliability of treatment effect inference.

In the experimental results, we report comparisons against two baseline models: the GRF and the DML estimator. Our proposed method is implemented within the econml Python package, which builds directly on the DML framework. As such, DML serves as the most relevant baseline for isolating the contribution of our enhancements to the training and tuning procedure. For completeness, numerical comparisons with both baselines (GRF and DML) are reported in the tables, while figures highlight the contrast with DML to better illustrate the gains of our proposed approach.

Table 1 summarizes the performance of all methods in terms of normalized uncertainty and normalized MSE across the Synthetic, Mindsets, and IHDP datasets. The uncertainty metric corresponds to the average width of the IJ-based CIs for the estimated CATEs, which reflects how precisely treatment effects are inferred. Both the CI width and MSE are divided by $\mathrm{Var}(Y)$, the variance of the outcome variable, to normalize for scale differences across datasets. This normalization enables results to be compared on a common basis, regardless of the original outcome range. Functionally, a lower normalized CI width indicates tighter confidence intervals and thus more reliable inference, while a lower normalized MSE indicates greater accuracy of treatment effect estimation. Viewed together, these two metrics characterize both the precision and accuracy of the competing models.

Across all datasets, the proposed method consistently delivers the tightest training and testing CIs-often reducing uncertainty by more than an order of magnitude compared to GRF and DML-while also achieving competitive or superior MSE performance. For example, in the Synthetic dataset with Lasso regressors, testing CI width decreases from 0.0517 (DML) to 0.0034 (a 93 % reduction), and testing MSE falls dramatically from 0.7927 to 0.0036. In the Mindsets dataset with RF regressors, our approach reduces testing CI width from 0.3780 (DML) to 0.0010 and simultaneously improves MSE relative to both baselines (0.6470 vs. 0.7630 and 0.7010). In the IHDP dataset, our method shrinks the testing CI width from 0.0135 to 0.0038 and improves MSE from 0.1121 to 0.0965. Taken together, these results show that the proposed method not only sharpens inference by producing substantially tighter confidence intervals but also generalizes better across train-test splits, as reflected in the smaller gaps between training and testing metrics. This combination of reduced estimation uncertainty and stable predictive accuracy underscores the reliability of our approach for treatment effect estimation across diverse data environments.

Although the normalized training MSE of the proposed model appears relatively higher than that of some baselines, this behavior is both expected and intentional. The model employs an honest data-splitting design in which disjoint subsets are used for tree construction, effect estimation, and hyperparameter tuning. This structure inherently constrains in-sample fit but preserves unbiasedness and prevents information leakage. Moreover, the composite optimization objective jointly minimizes prediction error and CI width, introducing an uncertainty-regularization effect that discourages overly complex tree structures. Consequently, the model deliberately trades a small amount of in-sample accuracy for tighter confidence intervals, reduced overfitting, and improved generalization. The close alignment between training and testing MSE values in Table 1 confirms that the slightly elevated training error reflects effective regularization rather than model misspecification. Additional performance metrics-including mean absolute error (MAE), root mean squared error (RMSE), and average confidence interval width (CI width)-are reported in Table D.1 within Appendix D. Moreover, complementary Taylor diagrams in Appendix D (Figs. D.4-D.6) visualize the joint relationships among correlation, normalized standard deviation, and predictive uncertainty, further confirming the superior calibration and accuracy of the proposed framework.

Fig. 5 presents the uncertainty of CATE estimation on the Mindsets dataset, comparing results from DML, GRF, and our proposed method. The vertical axis shows the estimated CATE with associated 95 % confidence intervals, while the horizontal axis represents the continuous covariate X_1, scaled to $[0, 1]$. The color scheme, blue for DML, orange for the proposed method, and green for GRF, clearly distinguishes the approaches. Relative to DML and GRF, our method achieves substantially tighter confidence intervals while maintaining comparable mean CATE estimates. This pattern holds across the covariate range, indicating that the gains are not confined to a narrow subset of X_1. The pronounced reduction in uncertainty highlights the effectiveness of the proposed approach in producing more stable and precise treatment effect estimates across the covariate range.

Fig. 6 compares the training and test losses (MSE, log scale) of DML and our proposed method across 100 realizations of the IHDP dataset. Each dot corresponds to a single realization, with dashed segments connecting the paired DML and proposed results. Panel (a) reports the training loss, where the proposed method consistently achieves smaller values than DML. On average across all realizations, the loss is reduced by 55.9 % (mean reduction), while in the typical

Table 1
Normalized IJ-based CI width and normalized MSE performance across three datasets (Synthetic, Mindsets, IHDP).

Dataset	$m(\cdot)$ and $e(\cdot)$	Method	Normalized Training Uncertainty	Normalized Testing Uncertainty	Normalized Training MSE	Normalized Testing MSE
Synthetic	**RF Regressors**	GRF	0.0108	0.0107	0.1900	0.1760
		DML	0.0454	0.0446	0.1640	0.8170
		Proposed	**0.0011**	**0.0004**	0.2550	0.2770
	Lasso Regressors	DML	0.0485	0.0517	0.8110	0.7927
		Proposed	0.0026	0.0034	**0.0035**	**0.0036**
Mindsets	**RF Regressors**	GRF	0.0108	0.0105	0.6700	0.7010
		DML	0.0388	0.3780	**0.3580**	0.7630
		Proposed	**0.0013**	**0.0010**	0.5130	**0.6470**
	Lasso Regressors	DML	0.0379	0.3570	0.6340	0.6610
		Proposed	0.0015	0.0014	0.6250	0.6520
IHDP	**RF Regressors**	GRF	0.0169	0.0149	0.0921	0.1302
		DML	0.0150	0.0135	**0.0129**	0.1121
		Proposed	**0.0054**	**0.0038**	0.0950	**0.0965**
	Lasso Regressors	DML	0.0141	0.0132	0.1097	0.1301
		Proposed	0.0105	0.0092	0.0952	0.0955

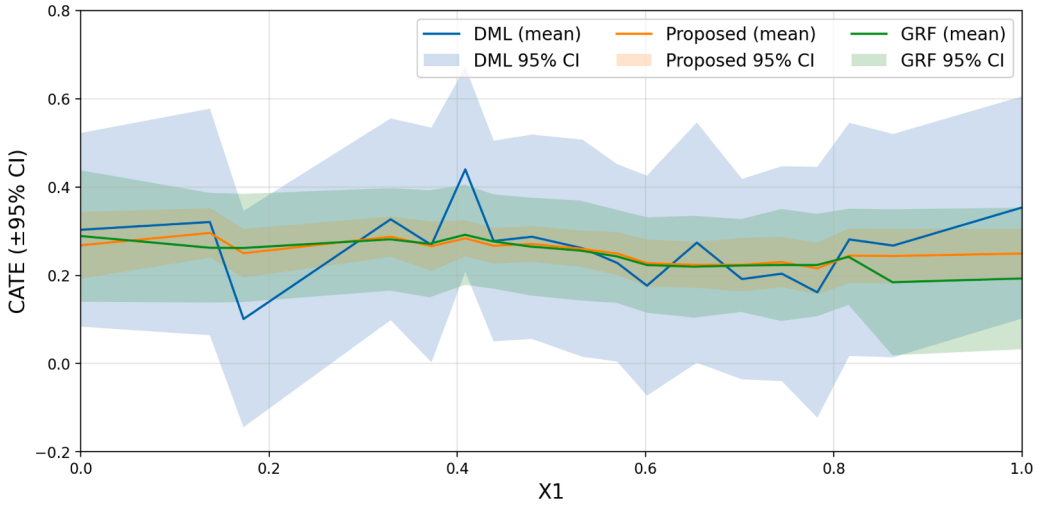


Fig. 5. Uncertainty of CATE estimation on Mindsets data.

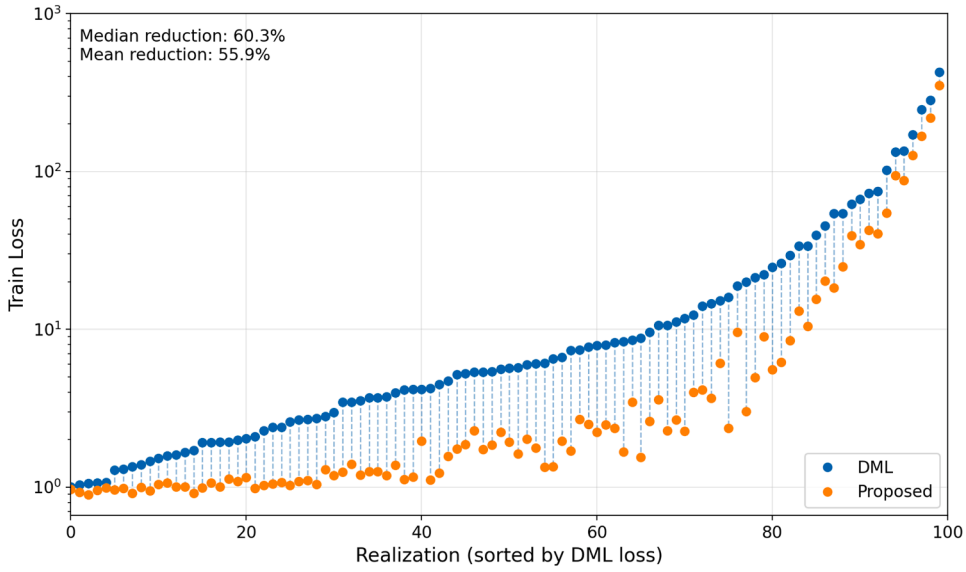
case-represented by the median-the reduction reaches 60.3 %. Panel (b) shows a similar pattern for test loss, with a mean reduction of 51.8 % and a median reduction of 56.7 %. The mean reduction quantifies the overall average improvement, whereas the median reduction highlights the typical or central tendency of improvement across realizations, making the results robust to outliers. Because each pair uses the same train/test split and seeds, improvements reflect model differences rather than split variability. The consistency of both statistics across training and testing confirms that our method not only improves in-sample fitting but also enhances generalization on held-out validation and test splits, yielding more robust and trustworthy treatment effect estimates.

Fig. 7 provides an alternative visualization of the training and test losses (MSE, log-log scale) for 100 IHDP realizations, comparing the proposed method with DML. Each point corresponds to one realization, with the dashed $y{=}x$ line indicating equal performance between the two methods. Points falling below this line represent realizations where the proposed method achieves lower loss than DML. As shown in panel (a) for training and panel (b) for testing, the vast majority of realizations lie below the diagonal, with median loss ratios of 0.40 and 0.43, respectively. These ratios imply that the proposed method reduces losses by roughly 60 % in training and 57 % in testing relative to DML. Together with the paired-difference plots, this scatterplot view reinforces the consistency and magnitude of the improvements, highlighting the effectiveness and robustness of the proposed framework in in-sample fitting and in generalization to held-out validation and test splits.Overall, these results corroborate that minimizing CI width during training does not sacrifice accuracy; instead, it coincides with lower error and improved stability.

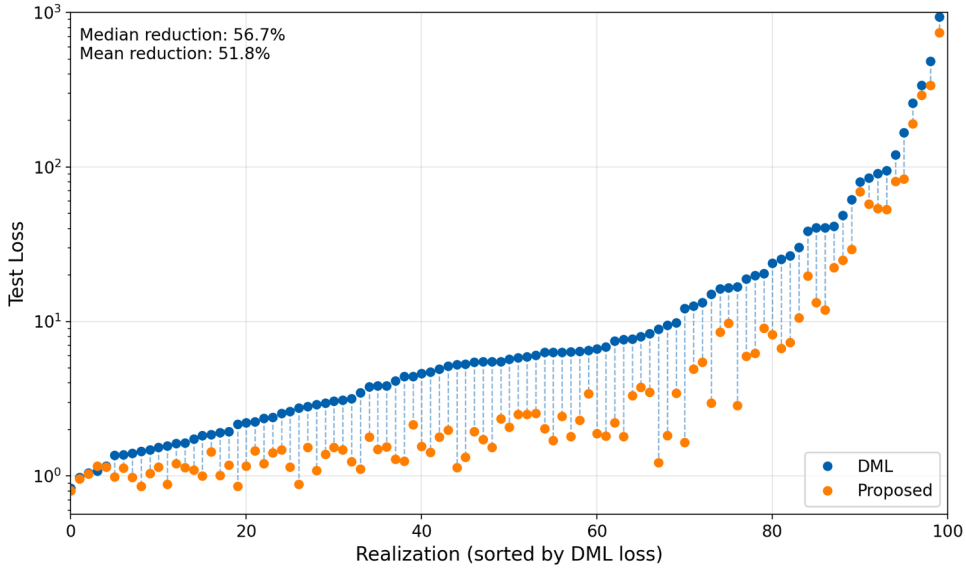
4.3.2. *Results of treatment allocation*

A central objective in many applications is to allocate costly treatments to those individuals most likely to benefit, provided that these benefits can be estimated with sufficient reliability. This targeting challenge arises in all three datasets studied-Synthetic, Mindsets, and IHDP. To illustrate, we focus here on the Job Training (Synthetic), where the improvements of our method are especially visible. In our targeting framework, individuals are prioritized when their estimated CATE exceeds the population mean and their 95 % confidence interval width is below the population mean. This dual criterion ensures that selected candidates are not only expected to gain more from treatment but that such gains are supported by precise, low-uncertainty estimates – i.e., finite-sample estimation uncertainty rather than irreducible outcome noise.

Fig. 8 compares the individuals highlighted by DML and by our proposed method. Panel (a) shows that the baseline DML approach produces highly variable CATE estimates with wide confidence intervals across education levels. As a result, the identified “high CATE & narrow



(a) Training loss across 100 IHDP realizations (DML vs. Proposed).



(b) Test loss across 100 IHDP realizations (DML vs. Proposed).

Fig. 6. Paired per-realization mean squared error (MSE, log scale) on the IHDP dataset. Each dot represents one realization, with dashed lines linking the corresponding DML and proposed losses. Lower values for the proposed method indicate improvement.

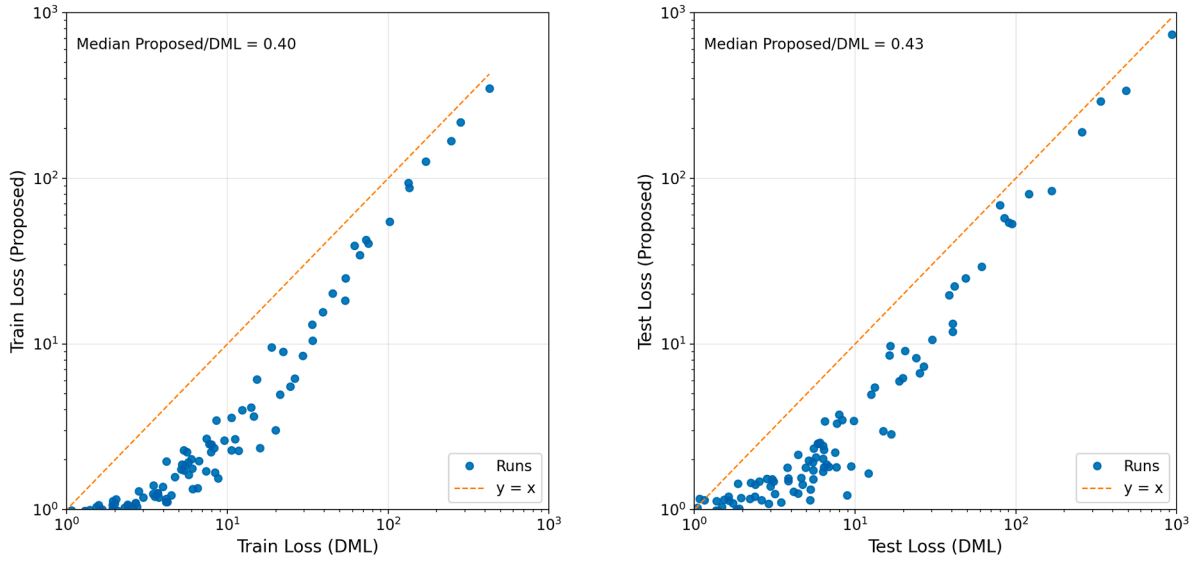
CI" candidates (yellow markers) are scattered loosely, and some may reflect noisy or unstable signals rather than reliable effects. In contrast, panel (b) demonstrates that our method yields smoother CATE trajectories with substantially narrower confidence intervals. The highlighted individuals are fewer and more concentrated, corresponding to regions where treatment effects are both sizable and statistically well supported.

This difference has direct implications for decision support. While DML risks over-selecting individuals based on unstable estimates, our framework produces a more trustworthy set of candidates, improving the efficiency and cost-effectiveness of treatment allocation. By reducing estimation uncertainty in the targeting process, the proposed method strengthens confidence that limited resources are directed to individuals who will genuinely benefit from intervention and naturally deprioritizes low-overlap regions where CI widths are large.

4.3.3. *Overfitting control and generalization results*

Table 1 reports the normalized uncertainty and MSE performance across the three datasets: Synthetic, Mindsets, and IHDP. The results show that our proposed method mitigates overfitting more effectively than both GRF and DML, consistently producing lower uncertainty values and maintaining close alignment between training and testing errors.

For the Synthetic dataset, the proposed method achieves the lowest uncertainty under both random forest and Lasso regressors. With RF Regressors, training uncertainty falls from 0.0108 (GRF) and 0.0454 (DML) to 0.0011, while testing uncertainty drops to 0.0004. Crucially, the training and testing MSE remain nearly identical (0.255 vs. 0.277), in contrast to DML, which exhibits a sharp discrepancy (0.164 vs. 0.817). A similar pattern holds for Lasso regressors, where our method reduces



(a) Training loss across 100 IHDP realizations.

(b) Test loss across 100 IHDP realizations.

Fig. 7. Scatterplots of DML vs. proposed MSE (log-log scale). Points below the dashed $y{=}x$ line indicate lower loss from the proposed method.

testing MSE from 0.7927 (DML) to 0.0036 while also lowering uncertainty. These patterns are consistent with the intended *uncertainty regularization* effect of our outer objective: minimizing CI width discourages overly complex trees that overfit idiosyncrasies of the training data.

Results on the Mindsets dataset reinforce this trend. With RF Regressors, testing uncertainty decreases dramatically from 0.3780 (DML) to 0.0010, and the gap between training and testing MSE narrows substantially (0.5130 vs. 0.6470). By comparison, DML produces a much lower training MSE (0.3580) but inflates the testing MSE (0.7630), a clear sign of overfitting. Under Lasso regressors, our method again maintains balanced training and testing errors while reducing uncertainty.

Finally, on the IHDP dataset, the proposed approach delivers the most stable performance. With RF Regressors, testing uncertainty is reduced from 0.0135 (DML) to 0.0038, while training and testing MSE remain nearly identical (0.0950 vs. 0.0965). With Lasso regressors, the method lowers uncertainty and produces virtually indistinguishable training and testing MSE (0.0952 vs. 0.0955), further confirming its robustness.

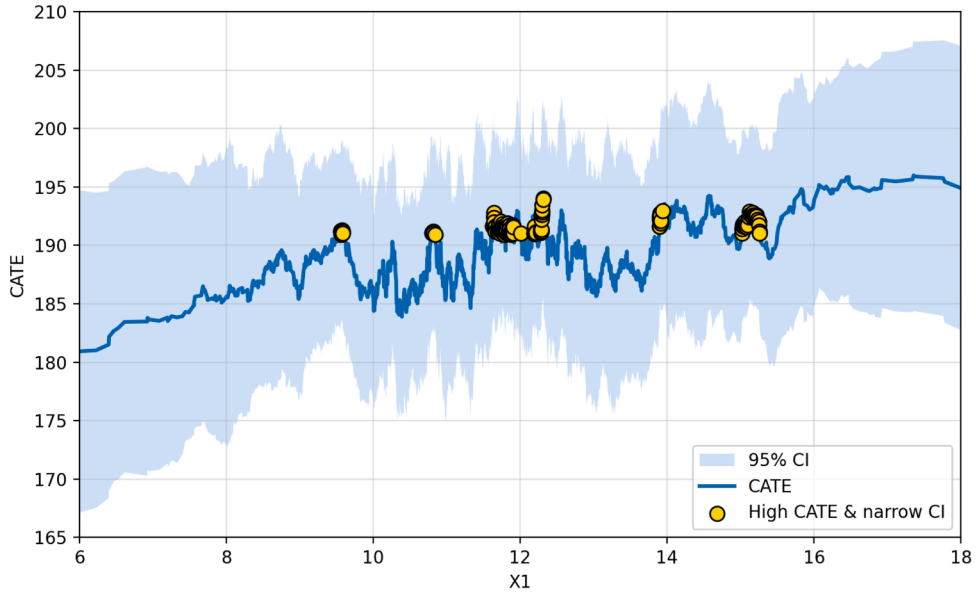
In summary, across all datasets and model specifications, our method not only reduces predictive uncertainty but also systematically minimizes the divergence between training and testing errors. These results highlight its effectiveness in controlling overfitting, thereby improving the reliability and generalizability of causal effect estimation. Practically, this means decision policies derived from our estimates are more likely to transfer faithfully to held-out validation and test settings. All code used for data preprocessing, model training, hyperparameter optimization, and evaluation is publicly available at the following GitHub repository: https://github.com/RAMSIS-Lab/Uncertainty_aware_causal_forest. The repository includes scripts for reproducing experiments, generating figures, and computing confidence intervals, along with detailed documentation for running the three-way nested validation framework.

5. Discussion

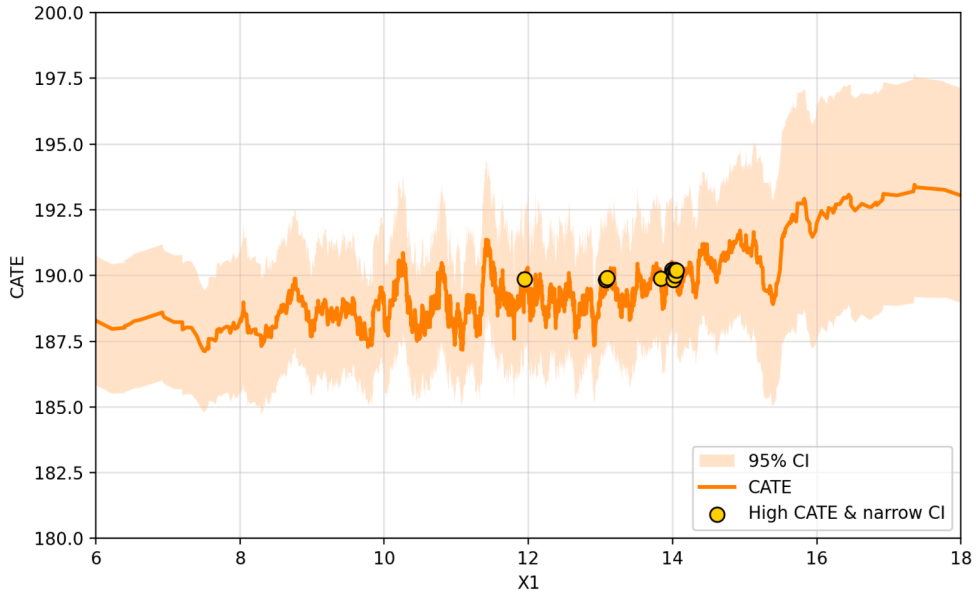
In the realm of data-driven decision-making, understanding causal relationships is crucial. This study introduces a framework that minimizes both estimation error and predictive uncertainty in heterogeneous treatment effect estimation by embedding Bayesian optimization into causal forests. Across synthetic and empirical datasets, the results demonstrate consistent improvements in predictive accuracy and uncertainty calibration relative to benchmark models, indicating that the joint optimization of mean squared error and confidence interval width yields more stable and interpretable treatment effect estimates.

This performance is achieved by advancing the literature on causal inference and CATE estimation through four main contributions. First, the proposed approach explicitly quantifies and minimizes the uncertainty of individualized treatment effects by incorporating *infinitesimal-jackknife (IJ)-based CI width* as a training objective. Second, the integration of Gaussian-process Bayesian optimization for joint hyperparameter tuning of accuracy and uncertainty provides a principled mechanism for balancing predictive and inferential quality. Third, the framework introduces a systematic cross-validation layer that enforces honest splitting and improves calibration, ensuring reliable and generalizable estimates. Finally, it proposes a treatment targeting rule that integrates both predicted treatment effects and their associated uncertainty, providing a theoretically grounded, cost-effective approach to intervention allocation.

Beyond methodological advancements, the framework also contributes to improving internal validity, external validity, and confounding control in causal machine learning. Internal validity is reinforced through the honest, nested cross-validation procedure, which strictly separates data used for model training, validation, and evaluation, thereby preventing information leakage and ensuring unbiased estimation within the observed sample. External validity is supported by the uncertainty-aware optimization that penalizes unstable estimates and by the dual-threshold targeting rule, which focuses inference on covariate regions with strong empirical support and adequate overlap. Potential confounding is mitigated by incorporating rich pre-treatment covariates in both the outcome and propensity models, and by monitoring confidence interval width as a diagnostic indicator of residual bias. While no observational method can completely eliminate hidden confounding, the framework provides transparent uncertainty measures that help identify regions where causal estimates are less reliable. Future extensions could enhance external generalization by integrating sample-reweighting or transportability adjustments with causal forest estimation, enabling treatment effects to be more robustly applied to target populations with differing covariate distributions.



(a) Targeted individuals selected by DML.



(b) Targeted individuals selected by the proposed method.

Fig. 8. Comparison of targeted individuals under DML and the proposed method for the Synthetic dataset. Points represent candidates with above-average CATE and below-average CI width.

From a managerial and policy perspective, the findings carry two key implications. First, organizations and policymakers are encouraged to adopt causal machine learning methods, such as the proposed framework, for individualized treatment effect estimation. Unlike traditional causal inference techniques that rely on strong parametric assumptions, causal machine learning flexibly models complex feature interactions and supports high-dimensional data. By explicitly quantifying the uncertainty of CATE estimates, the proposed approach enhances the transparency and credibility of model outputs, fostering greater trust and collaboration between data scientists and domain experts. Second, the dual-threshold treatment targeting rule serves as a practical decision aid for allocating scarce or costly interventions. By prioritizing individuals who are expected to yield above-average treatment effects and exhibit below-average uncertainty, managers can achieve more effective and equitable resource allocation. The LCB/knapsack extension further enables budget-constrained or fairness-aware deployment, translating methodological improvements into actionable strategies.

An important consideration in model reliability is the potential for overfitting-when models capture idiosyncrasies of the training data rather than generalizable patterns. In this study, overfitting was systematically detected and controlled through a multi-layered design. Detection relied on comparing performance across training, validation, and testing sets within a three-way nested, honest cross-validation scheme, using both normalized MSE and IJ-based CI width as diagnostic metrics. Minimal train-test gaps observed in Table 1 and Figs. 6 to 7 indicate that the proposed framework achieves nearly identical in-sample

performance and generalization to held-out validation and test splits, whereas benchmark models (e.g., DML) exhibit substantial divergence. Control was achieved through multiple safeguards: (i) honest data splitting that prevents information leakage between model training and evaluation; (ii) bi-objective Bayesian optimization that regularizes model complexity by jointly minimizing MSE and CI width, penalizing high-variance or overly complex configurations; and (iii) convergence monitoring across optimization iterations to ensure stable generalization to held-out validation and test splits. Together, these mechanisms provide both empirical and procedural evidence that overfitting is effectively detected and mitigated, yielding models that generalize reliably across heterogeneous datasets.

The framework offers several clear advantages: it (i) enables direct optimization of uncertainty alongside accuracy, (ii) yields interpretable and well-calibrated CIs, and (iii) supports generalization across diverse data environments through principled Bayesian tuning and honest validation. However, these benefits come with trade-offs. The integration of Bayesian optimization and uncertainty-aware objectives increases computational complexity, particularly in high-dimensional or large-scale settings. The reliability of inference depends on accurate uncertainty estimation and the specification of Gaussian-process priors. Moreover, the dual-threshold targeting rule requires domain-specific calibration to align with practical cost and risk preferences. Finally, the current framework operates in a static context; extending it to online or dynamic environments remains an important area for future research. Acknowledging these limitations provides direction for improving scalability and adaptability in subsequent work. Importantly, the proposed cross-validation and uncertainty-minimization scheme is fully modular: it can be paired with alternative CATE estimators-such as meta-learners, conformal inference approaches, or neural-network-based models-without modification to the overall workflow. This flexibility enables subsequent studies to embed the same principled uncertainty control within more expressive or domain-specific estimators, broadening the applicability of the framework.

Beyond methodological development, the proposed framework can be directly integrated into expert decision-support systems. Its individualized effect estimates and associated confidence intervals can function as decision variables in modules that rank, prioritize, or allocate limited resources under uncertainty. For example, in healthcare or social policy applications, the Uncertainty-Aware Causal Forest framework can identify individuals who are expected to experience meaningfully positive treatment effects while also exhibiting low uncertainty in those estimates, thereby supporting transparent, risk-aware decisions. This integration aligns the framework with the practical needs of expert systems that must balance accuracy, interpretability, and confidence in automated recommendations.

While this study focused on established benchmark datasets to ensure reproducibility and ground-truth validation, the proposed framework is directly transferable to real-world expert systems. In domains such as healthcare, customer analytics, and energy management, the ability to quantify and minimize uncertainty in individual treatment effects can improve resource allocation and policy targeting. Future applications will focus on deploying and evaluating the framework in large-scale operational datasets to assess its robustness under real-world complexities.

6. Conclusion

This paper proposed an *Uncertainty-Aware Causal Forest* framework that jointly optimizes predictive accuracy and estimation uncertainty to support robust treatment assignment in data-driven decision systems. By integrating multi-objective Bayesian optimization, explicit cross-validation, and a dual-threshold targeting rule, the framework advances both methodological rigor and practical applicability. Empirical evaluations across synthetic and benchmark datasets demonstrate that the proposed method produces tighter and better-calibrated confidence intervals while maintaining competitive predictive accuracy relative to established baselines. These results contribute to the growing class of uncertainty-aware intelligent systems that enhance the reliability, interpretability, and transparency of machine-learning-based decision support. The proposed framework complements broader advances in intelligent system design that emphasize robustness, regularization, and transparency in predictive modeling. By explicitly minimizing estimation uncertainty, it extends these principles to the causal-inference domain, enabling interpretable and risk-aware policy recommendations across applications such as healthcare, finance, and energy management. In doing so, it bridges methodological innovation with the practical needs of expert systems that must balance predictive performance with decision confidence.

Looking ahead, several research directions remain open. First, future work could extend the proposed framework to large-scale, dynamic decision-support environments such as healthcare triage systems or energy market policy design, where uncertainty-aware treatment targeting has tangible real-world implications. Second, developing adaptive or online versions of the Uncertainty-Aware Causal Forest would enable real-time learning and continual model refinement under streaming data. Third, integrating domain-specific cost structures or fairness constraints into the joint optimization could further align the method with ethical and operational considerations in high-stakes decision-making.

CRediT authorship contribution statement

Yufan Ji: Conceptualization, Methodology, Investigation, Visualization, Validation, Writing – original draft; **Abdollah Shafieezadeh:** Conceptualization, Methodology, Validation, Supervision, Writing – review & editing.

Data availability

The codes and models are provided via GitHub. The link is referenced in the manuscript

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

Supplementary material associated with this article can be found in the online version at 10.1016/j.eswa.2025.130578

Appendix A. Visualization and diagnostic checks for input features and covariate balance

To support the credibility of the causal analysis, this appendix presents exploratory visualizations and balance diagnostics for all three datasets used in the study. These include univariate feature distributions by treatment status and propensity score overlap plots.

A.1. Job training dataset

Fig. A.1 presents exploratory visual diagnostics for the Job Training dataset. Panel (a) shows a strong positive relationship between education level and post-training earnings, with treated individuals generally achieving higher earnings across the education spectrum. Panel (b) indicates that the treatment assignment is not systematically related to age, as both groups are broadly distributed over similar age ranges. Panel (c) compares the overall earnings distributions, showing that while treated participants tend to earn more on average, the two groups exhibit substantial overlap. Together, these patterns confirm that the dataset exhibits balanced yet informative variation suitable for causal analysis.

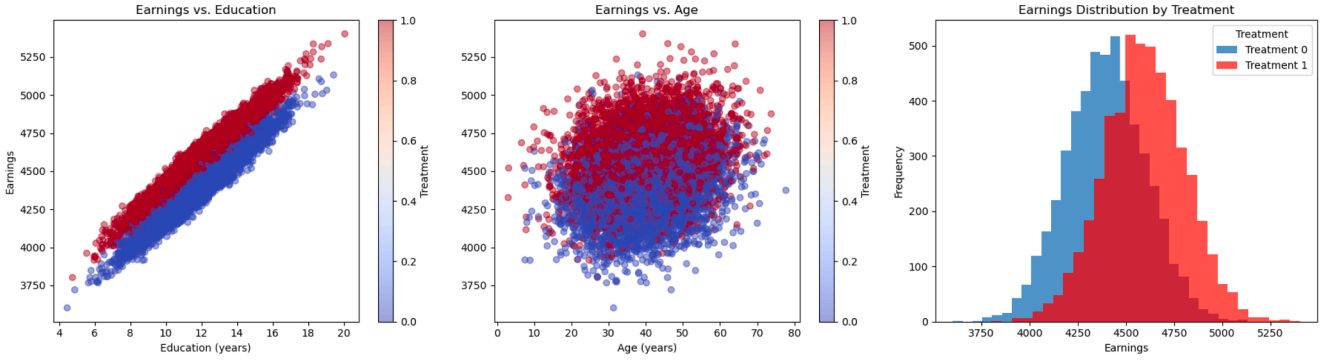


Fig. A.1. Visual diagnostics of input features in the Job Training dataset. The left and middle panels show scatter plots of *Earnings* versus *Education* and *Age*, respectively, colored by treatment status. The right panel presents the marginal distribution of *Earnings* for treated and control groups.

A.2. Learning mindset dataset

Figs. A.2 and A.3 summarize the covariate distributions and treatment assignment balance for the Learning Mindset dataset. Fig. A.2 shows that treated and control groups exhibit modest distributional differences across several covariates, indicating some degree of selection into treatment. Fig. A.3 illustrates that the estimated propensity scores display strong overlap between the two groups, suggesting adequate common support for causal inference. Together, these diagnostics confirm that the dataset satisfies key covariate balance and overlap conditions required for unbiased estimation of treatment effects. Detailed definitions of the covariates are provided in Athey and Wager (2019).

A.3. Infant health and development program (IHDP) dataset

Fig. A.4 presents pairwise scatter and density plots for selected covariates and the outcome in the IHDP dataset, stratified by treatment status. The relationships among X_1-X_3 show clear but overlapping differences between treated and control groups, suggesting moderate selection into treatment alongside meaningful variation across units. The outcome distributions reveal higher values for treated observations, consistent with the dataset's known positive treatment effect. Overall, the figure demonstrates that the IHDP data contain both sufficient heterogeneity and overlap to enable reliable estimation of causal effects. Detailed descriptions of variables and data collection procedures are available in the original IHDP documentation and cited references.

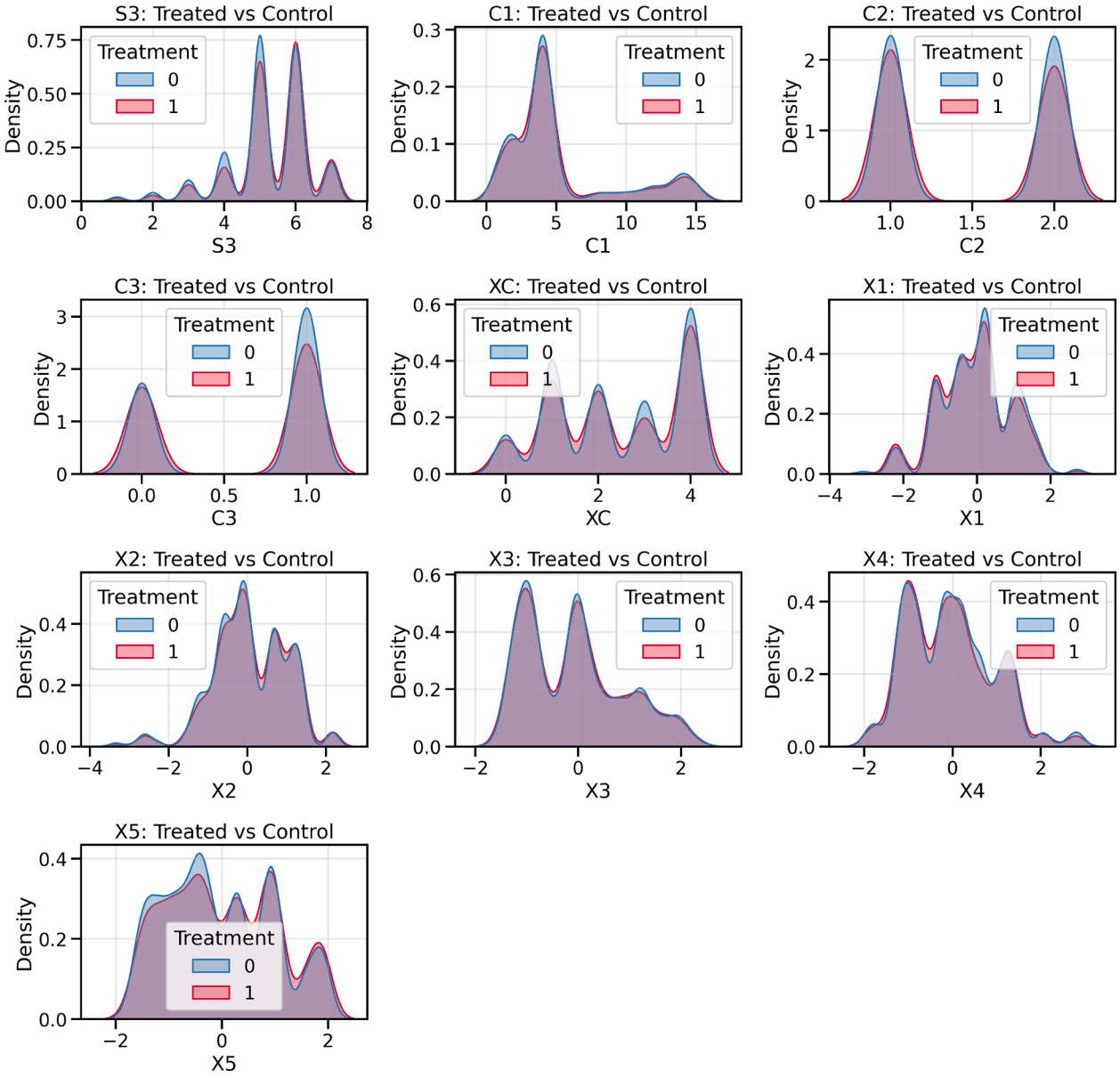


Fig. A.2. Distributions of input covariates by treatment status in the Learning Mindset dataset. Each panel compares the density of a covariate between treated and control groups.

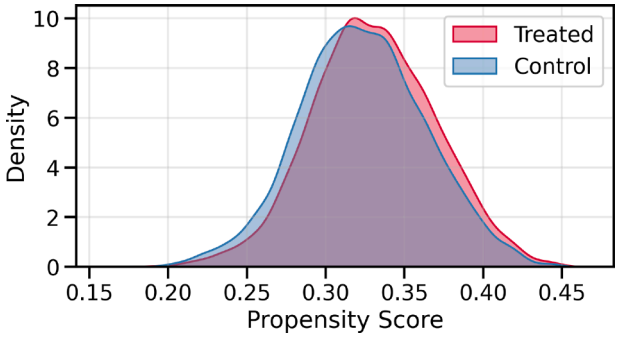


Fig. A.3. Propensity score distributions for treated and control groups in the Learning Mindset dataset.

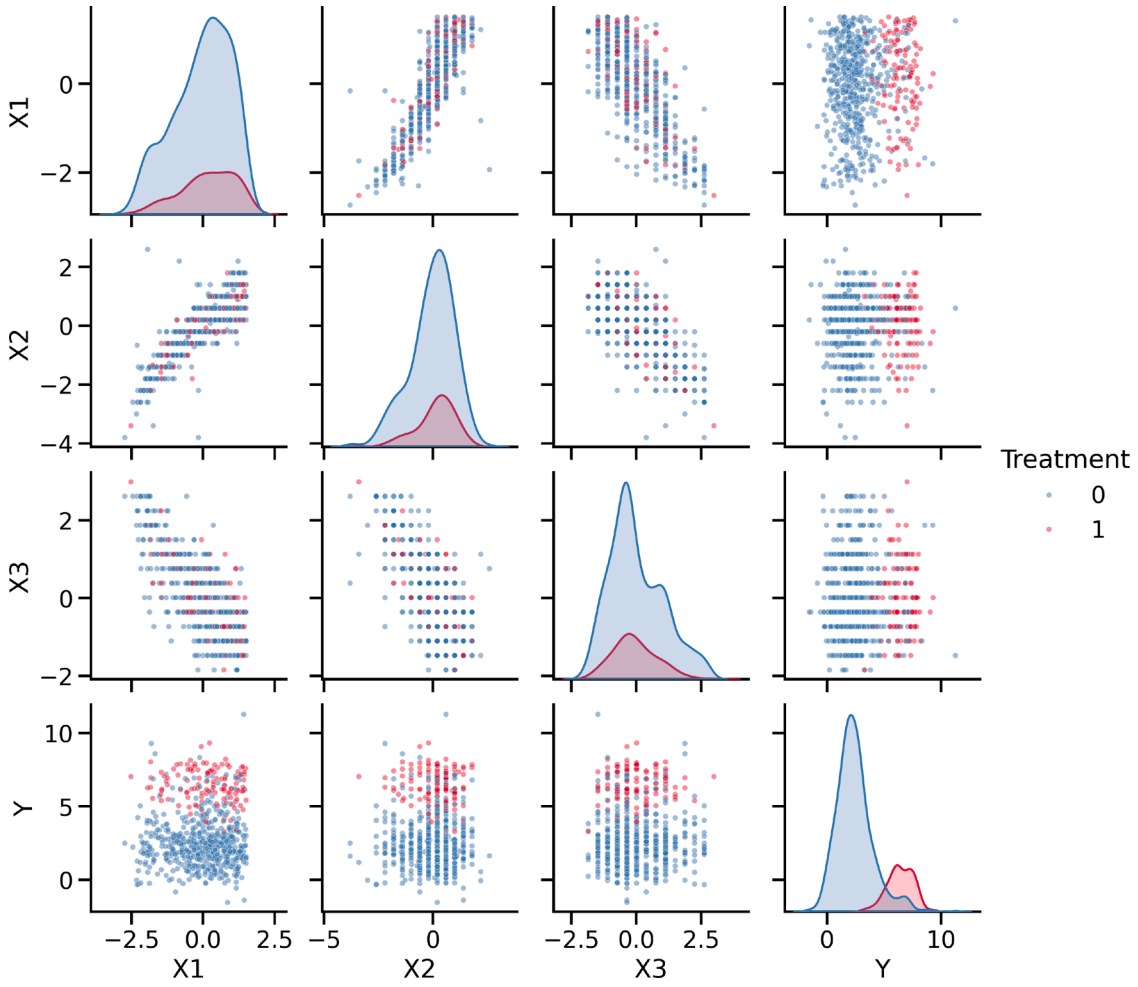


Fig. A.4. Selected features versus outcome by treatment status in the IHDP dataset. Each panel presents pairwise relationships among a subset of covariates (X_1-X_3) and the outcome variable (Y), with treated and control observations shown in different colors.

Appendix B. Hyperparameter space and justification

Table B.1 provides details of the hyperparameter space, search ranges, and justification for the Bayesian optimization process described in Section 3.2. All hyperparameters are tuned jointly within the two-layer optimization framework, where the inner layer minimizes predictive error (MSE) and the outer layer minimizes confidence interval width. The selected hyperparameters control the structural complexity, regularization strength, and stability of the causal forest.

The search ranges were guided by prior literature on causal forests and generalized random forests (ATHEY et al., 2019; Wager & Athey, 2018), which theoretically characterize how tree depth, leaf size, and sample partitioning govern the bias-variance trade-off and asymptotic consistency. The specific numeric ranges were further informed by empirical studies and practical implementations of causal forest models in applied econometric and policy contexts (Athey & Imbens, 2016; Künzel et al., 2019; Tibshirani et al., 2024), as well as by preliminary sensitivity analyses that ensured stable variance estimates without excessive computational burden. Bayesian optimization was conducted using Gaussian process surrogates with an acquisition function that adaptively balances exploration and exploitation across the hyperparameter space. The optimal configuration $\phi^\star$ was obtained by minimizing the composite objective in Eq. (14) using fivefold cross-validation.

Table B.1
Hyperparameter space, search ranges, and justification for Bayesian optimization of the causal forest.

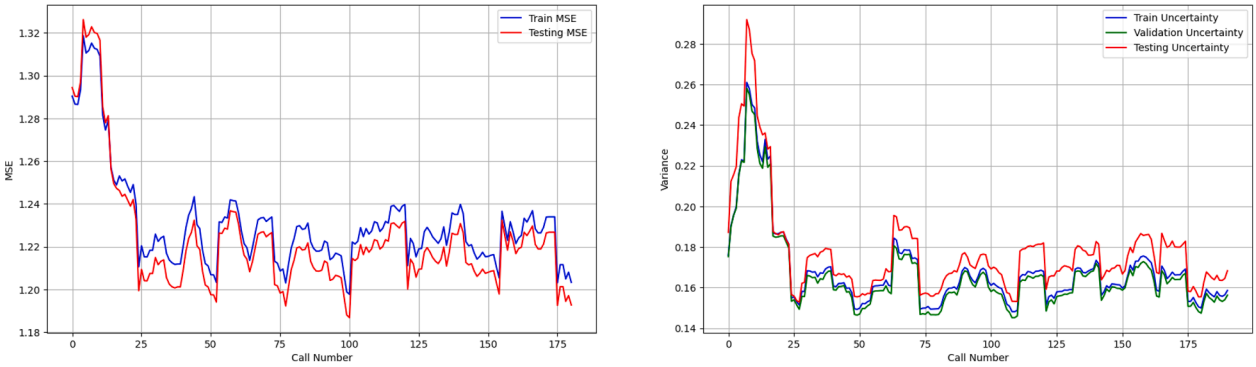
Hyperparameter	Type	Search Range	Justification / Role
`n_estimators` (number of trees)	Integer	[50, 500]	Controls ensemble size; larger forests reduce variance by averaging across more trees but increase computational cost. A moderate upper bound ensures stability without overfitting.
`max_depth` (maximum tree depth)	Integer	[3, 15]	Limits the depth of each tree, constraining model complexity and preventing overfitting. Deeper trees capture more heterogeneity but may inflate variance in small samples.
`min_samples_split` (minimum samples to split)	Integer	[2, 20]	Determines the minimum number of observations required to perform an internal split. Higher values enforce smoother partitions and reduce sensitivity to noise.
`min_samples_leaf` (minimum leaf size)	Integer	[2, 50]	Specifies the minimum number of observations allowed in each terminal leaf. Larger values stabilize local CATE estimates and mitigate high variance in leaf-level treatment effects.

Appendix C. Quantitative validation of joint optimization

In this appendix, a quantitative assessment of the proposed joint optimization framework is presented through three complementary analyses: (i) empirical convergence behavior, (ii) bias–variance trade-off, and (iii) uncertainty calibration. All experiments are performed on the Job Training (synthetic) dataset. Within the causal forest, Lasso regressors with fivefold cross-validation are employed as nuisance models for both the outcome model $\hat{m}$ and the propensity score model $\hat{e}$.

C.1. Empirical convergence analysis

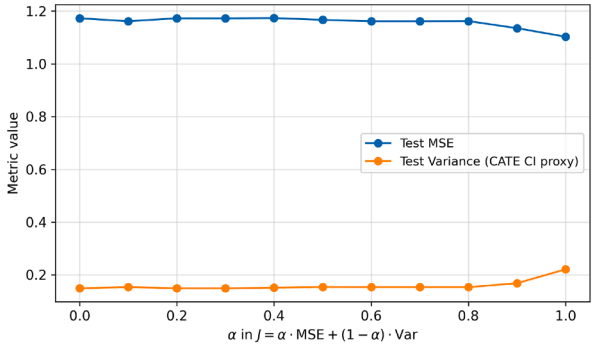
Fig. C.1 provides a quantitative evaluation of the convergence behavior during the Bayesian optimization process. Panel (a) shows that both training and testing mean-squared errors (MSE) decrease sharply in the early iterations and then plateau, indicating that the model rapidly reaches a stable optimum with respect to predictive accuracy. Panel (b) reports the corresponding evolution of predictive uncertainty, expressed as the variance of estimated confidence interval widths for the conditional average treatment effect (CATE). The uncertainty curves for the training, validation, and testing sets exhibit nearly identical shapes, each declining until about the 25th call and stabilizing thereafter, demonstrating empirical convergence of the uncertainty objective as well. The close alignment among all three sets indicates that the optimization generalizes well across folds and avoids



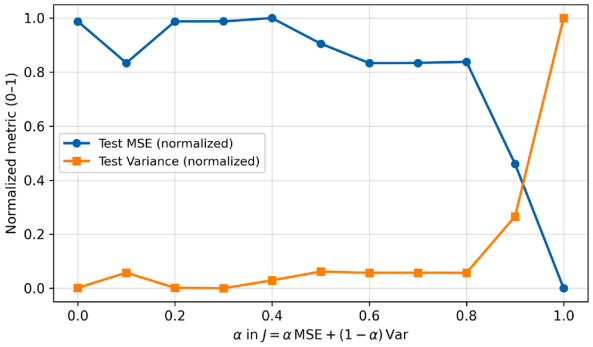
(a) Evolution of training and testing MSE across Bayesian optimization calls.

(b) Evolution of training, validation, and testing uncertainty (variance proxy) across calls.

Fig. C.1. Empirical convergence of the proposed joint optimization framework on the Job Training dataset using Lasso regressors within the causal forest.



(a) Raw MSE and variance values across α settings.



(b) Normalized MSE and variance on a common 0–1 scale.

Fig. C.2. Bias-variance trade-off analysis of the proposed Uncertainty-Aware Causal Forest on the Job Training dataset using Lasso regressors for both outcome and treatment models. The joint objective $J = \alpha\,\mathrm{MSE} + (1-\alpha)\,\mathrm{Var}$ balances predictive accuracy and uncertainty reduction through Bayesian optimization.

overfitting. Together, these results provide strong evidence that the proposed joint optimization framework achieves stable convergence in both accuracy and uncertainty dimensions.

C.2. Bias–variance trade-off analysis

Figs. C.2 and C.3 collectively illustrate how the proposed framework balances predictive accuracy and estimation uncertainty through the weighting parameter α, which controls the relative importance of the mean-squared error (MSE) and predictive variance in the joint objective Eq. (14).

In Fig. C.2, Panel (a) shows that as α increases, the optimization increasingly prioritizes predictive accuracy, yielding lower MSE values but slightly higher predictive variance. Panel (b) presents the same results on a normalized scale, which highlights the inverse relationship between the two objectives: emphasizing uncertainty minimization ($\alpha \approx 0$) produces narrower confidence intervals at the cost of marginally higher prediction error, whereas larger α values reduce MSE while tolerating greater uncertainty. The consistent and monotonic patterns observed across both panels confirm that the proposed Uncertainty-Aware Causal Forest achieves a smooth and interpretable bias–variance transition, allowing users to tune α according to desired trade-offs between precision and reliability.

Complementing this analysis, Fig. C.3 depicts the Pareto frontier summarizing the optimal combinations of MSE and variance identified through Bayesian optimization. Each point represents the best-performing configuration for a specific α in the joint objective (14). The downward-sloping frontier demonstrates a well-defined and stable trade-off: models with lower predictive variance (tighter CATE confidence intervals) tend to incur slightly higher MSE, while those emphasizing accuracy achieve smaller errors at the expense of increased uncertainty. This smooth Pareto-efficient curve confirms that the proposed joint optimization framework effectively spans the space between accuracy- and uncertainty-oriented regimes, providing practitioners with a transparent mechanism to select operating points consistent with their decision-making goals and acceptable uncertainty levels.

C.3. Uncertainty calibration analysis

Table C.1 summarizes the uncertainty calibration performance of the proposed framework using conformal prediction applied to the estimated CATEs. For each nominal confidence level, we report the corresponding empirical pseudo-coverage and the average width of the conformal prediction intervals. The pseudo-coverage metric provides a practical means of assessing interval calibration when ground-truth treatment effects are unavailable. Specifically, for each observation in the held-out test set, conformal prediction intervals for the estimated treatment effect $\hat{\tau}_i$ are constructed using the proposed Uncertainty-Aware Causal Forest. To approximate the unobserved true effects, we compute a *doubly robust (DR) pseudo-label* for each unit:

$$\tilde{\tau}_i = (T_i - \hat{e}(X_i))\,\frac{Y_i - \hat{m}(X_i)}{\hat{e}(X_i)\,(1-\hat{e}(X_i))}, \tag{C.1}$$

where $\hat{m}(X_i)$ and $\hat{e}(X_i)$ denote the estimated outcome and propensity models, respectively. This DR estimator combines outcome and treatment residuals to yield an asymptotically unbiased proxy for the individual treatment effect. The empirical pseudo-coverage is then computed as the proportion of test observations whose DR pseudo-labels fall within their respective conformal intervals, i.e.,

$$\text{Pseudo-coverage} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\big(\tilde{\tau}_i \in [\hat{\tau}_i^L, \hat{\tau}_i^U]\big), \tag{C.2}$$

which provides a consistent approximation of interval validity even when the true treatment effects are not directly observable.

As shown in Table C.1, the empirical pseudo-coverages (0.791, 0.903, and 0.954) align closely with their corresponding nominal levels (0.80, 0.90, and 0.95), indicating that the constructed intervals are well-calibrated and maintain appropriate coverage across multiple confidence levels. Meanwhile, the average interval widths increase smoothly with higher confidence levels (from 5.51 to 8.67), reflecting consistent scaling of uncertainty rather than over- or underestimation. These results demonstrate that the proposed Uncertainty-Aware Causal Forest yields reliable and interpretable uncertainty quantification, producing confidence intervals that are both accurate in coverage and adaptive in width, thereby enhancing the transparency and credibility of treatment effect estimation.

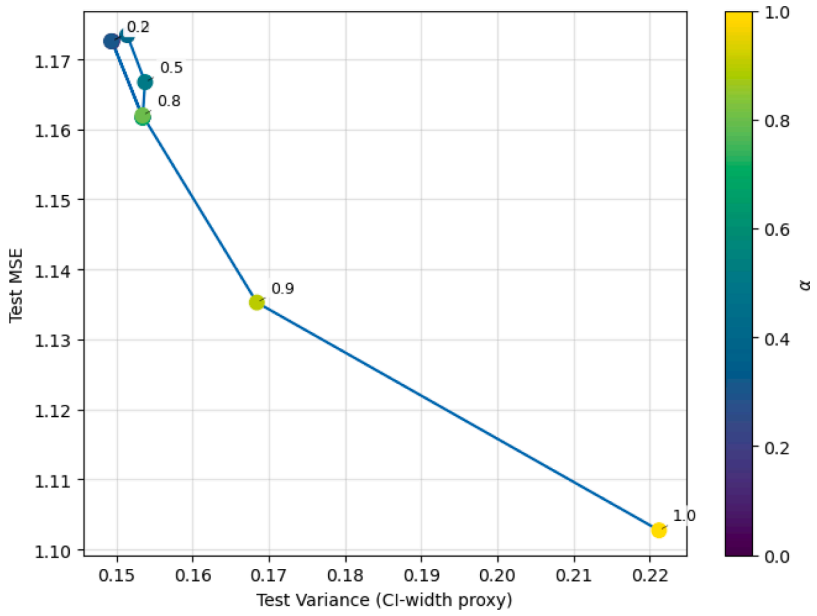


Fig. C.3. Pareto frontier between predictive accuracy and uncertainty for the proposed joint optimization framework on the *Job Training (synthetic)* dataset. Each point represents the best-performing configuration for a specific weighting parameter α in the joint objective $J = \alpha\,\mathrm{MSE} + (1-\alpha)\,\mathrm{Var}$. The color gradient encodes increasing values of α from 0 (purple) to 1 (yellow).

Table C.1
Conformal uncertainty calibration results.

Conformal Level	Pseudo-Coverage	Average Width
0.80	0.791	5.506
0.90	0.903	7.275
0.95	0.954	8.666

Appendix D. Additional performance diagnostic

Table D.1 reports the generalization performance on held-out validation and test splits of the benchmark models (GRF, DML) and the proposed framework across the three datasets. For each model, the Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and average confidence interval (CI) width are computed on held-out test sets. The MAE and RMSE values are calculated by comparing the predicted outcomes $\hat{Y}$—obtained from the fitted outcome and propensity models within each framework—against the observed outcomes Y. This comparison of generalization to held-out validation and test splits provides a direct measure of predictive fidelity under the honest cross-validation structure, quantifying how accurately the models reproduce observed outcomes while maintaining strict data separation. The average CI width reflects the precision of the estimated conditional average treatment effects (CATEs), with narrower intervals indicating lower estimation uncertainty.

Consistent with the main results presented in Section 4.3, the proposed framework achieves the lowest MAE, RMSE, and average CI width in nearly all cases, demonstrating superior predictive accuracy and tighter uncertainty quantification. For the Job Training and Learning Mindsets datasets, the improvements are particularly pronounced, reflecting enhanced generalization and stability of the estimated CATEs. In the more challenging IHDP setting, the proposed model continues to reduce RMSE relative to both baselines and yields narrower confidence intervals, suggesting more efficient variance estimation. Overall, these results reinforce that the proposed framework provides a well-balanced trade-off between accuracy and uncertainty, outperforming generalized random forest (GRF) and double machine learning (DML) estimators across diverse data environments.

Figs. D.4–D.6 present Taylor diagrams summarizing the relationship between correlation, normalized standard deviation, and predictive uncertainty for each dataset. The proposed framework consistently achieves higher correlation with the observed outcomes and maintains a normalized standard deviation closer to the reference arc ($\hat{\sigma}/\sigma_o = 1$), indicating improved calibration and stability of the estimated effects. In both the Job training and Learning Mindsets datasets, the proposed model aligns most closely with the observed reference, showing superior accuracy and reduced dispersion compared to GRF and DML. For the more challenging IHDP dataset, although all models exhibit lower correlations, the proposed method still outperforms the baselines by achieving a better balance between correlation and variance.

Table D.1
Testing-set performance: mean absolute error (MAE), root mean squared error (RMSE), and average CI width. Boldface indicates the best (smallest) value within each dataset.

Dataset	Model	MAE	RMSE	CI width
Job training	GRF	6.2408	9.9126	8.3375
	DML	1.8277	2.2876	2.2775
	Proposed	**0.8528**	**1.0829**	**1.3580**
Learning Mindsets	GRF	0.4301	0.5388	0.2491
	DML	0.5034	0.6248	0.4749
	Proposed	**0.4158**	**0.5207**	**0.0970**
IHDP	GRF	9.4824	18.6673	**0.4144**
	DML	12.1110	19.7913	3.9549
	Proposed	**9.2409**	**15.9935**	2.0050

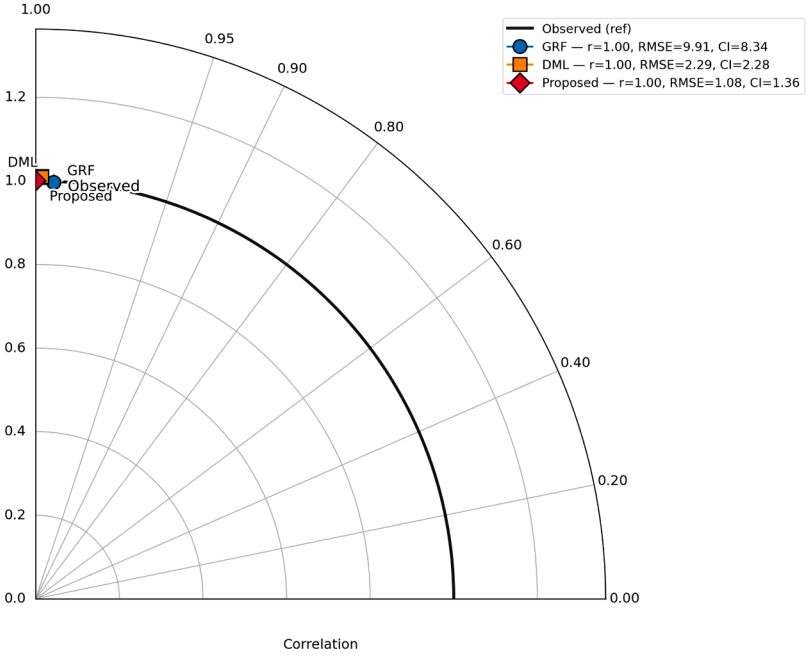


Fig. D.4. Taylor diagram (quarter view, normalized by σ_o) for the *Job training* dataset. The black arc at radius 1 marks $\hat{\sigma}/\sigma_o = 1$ (Observed reference). Legends report correlation r, RMSE, and CI width.

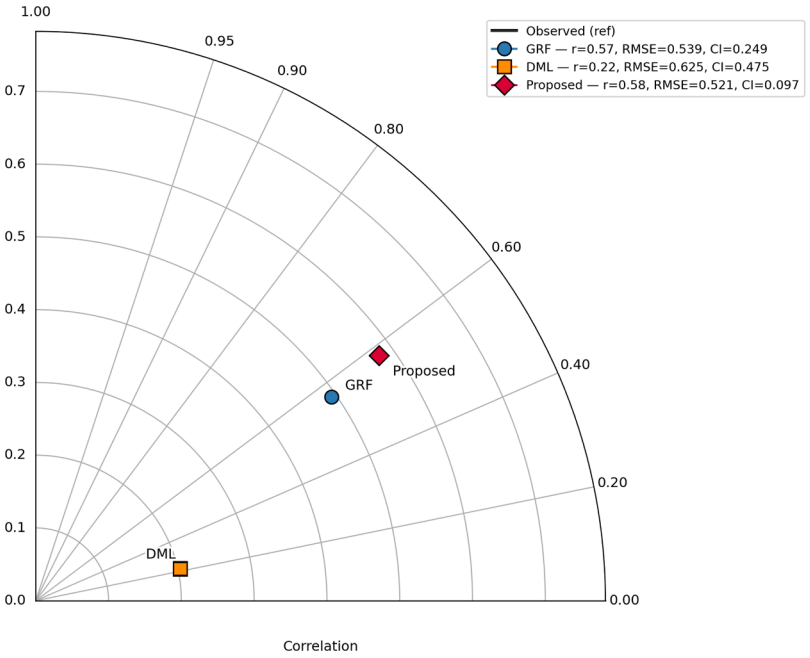


Fig. D.5. Taylor diagram (quarter view, normalized by σ_o) for the *Learning Mindsets* dataset. The black arc at radius 1 marks $\hat{\sigma}/\sigma_o = 1$ (Observed reference). Legends report correlation r, RMSE, and CI width.

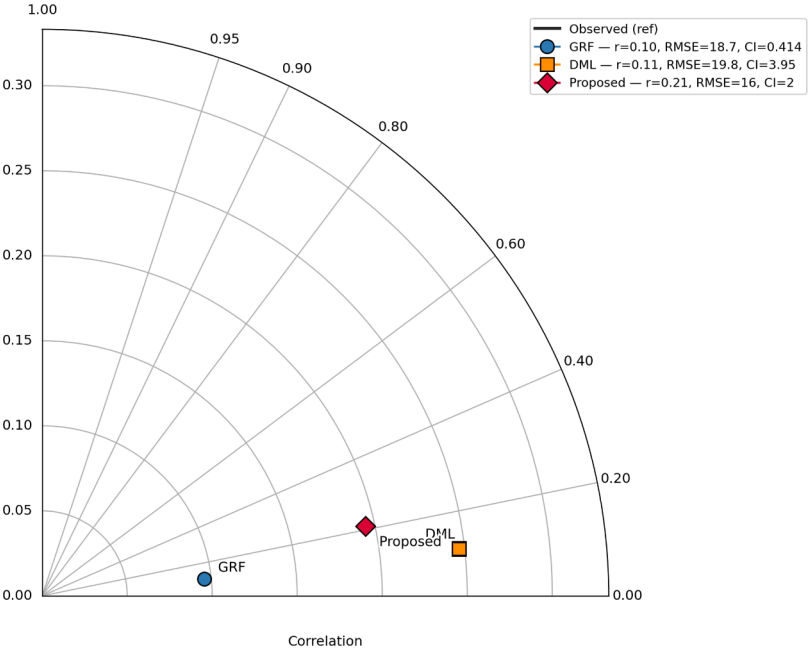


Fig. D.6. Taylor diagram (quarter view, normalized by σ_o) for the *IHDP* dataset. The black arc at radius 1 marks $\hat{\sigma}/\sigma_o = 1$ (Observed reference). Legends report correlation r, RMSE, and CI width.

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